LECTURE 9

NON-REVERSIBLE PARALLEL TEMPERING

Saifuddin Syed

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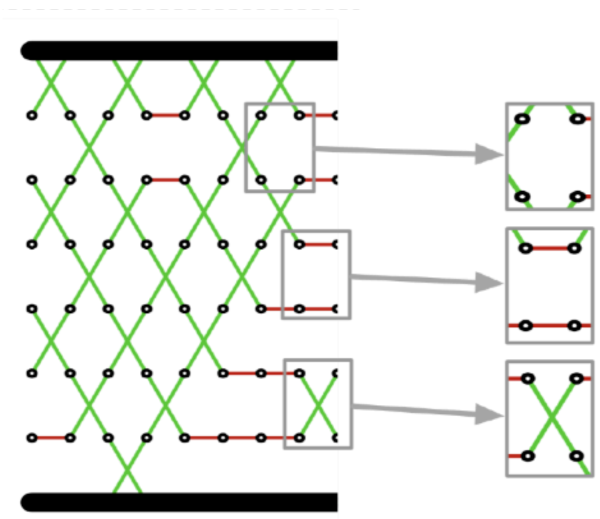

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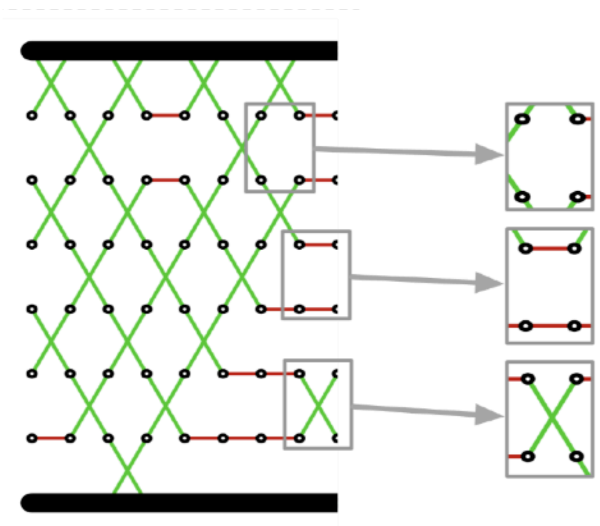

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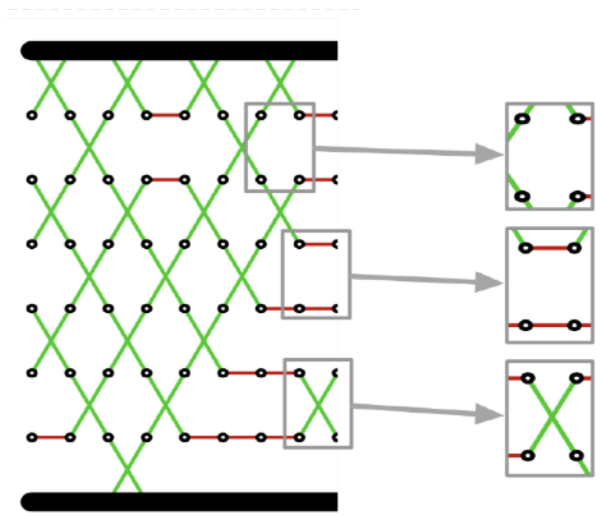

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- Due to complex interactions it is hard to analyse in general

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- We want to work with the stationary acceptance probability which are independent of $\mathbf{X}_t$

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3. **Efficient local exploration (ELE):** For all β if $X' \,|\, X \sim K_\beta(X, \mathrm{d}x')$ then $V(X') \perp V(X)$

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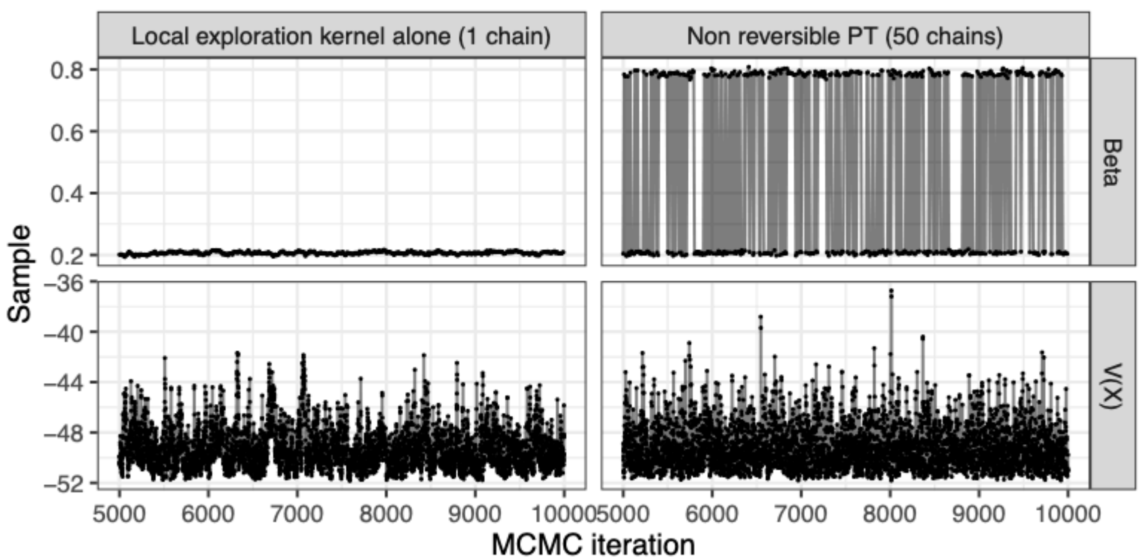

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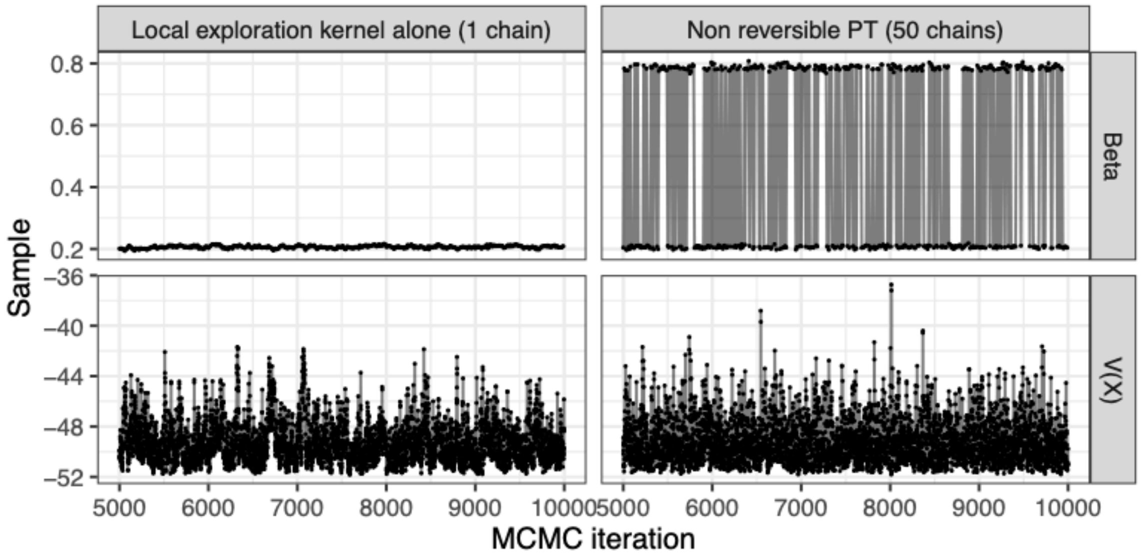
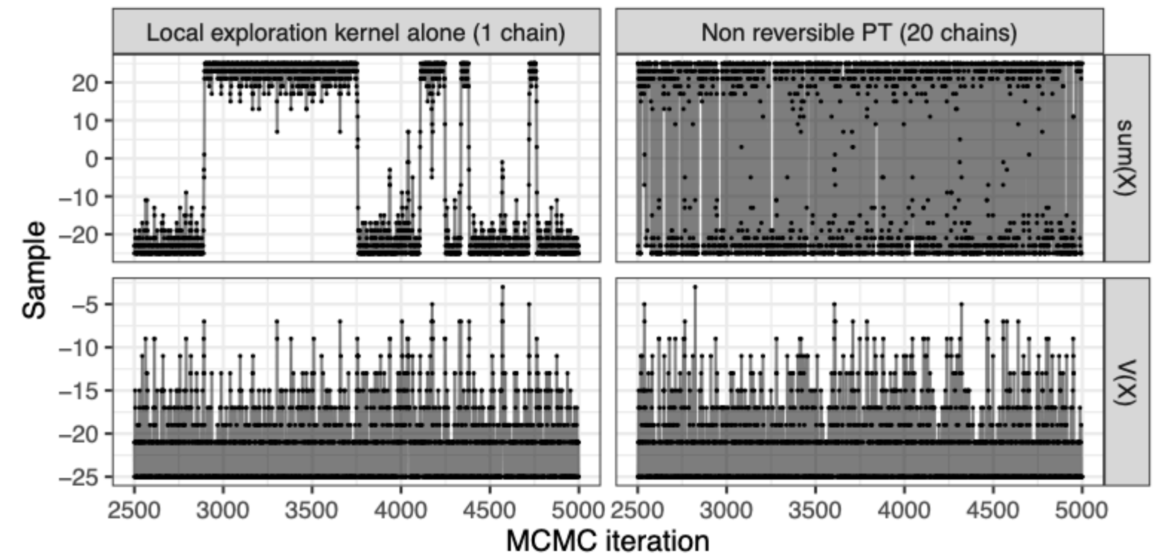
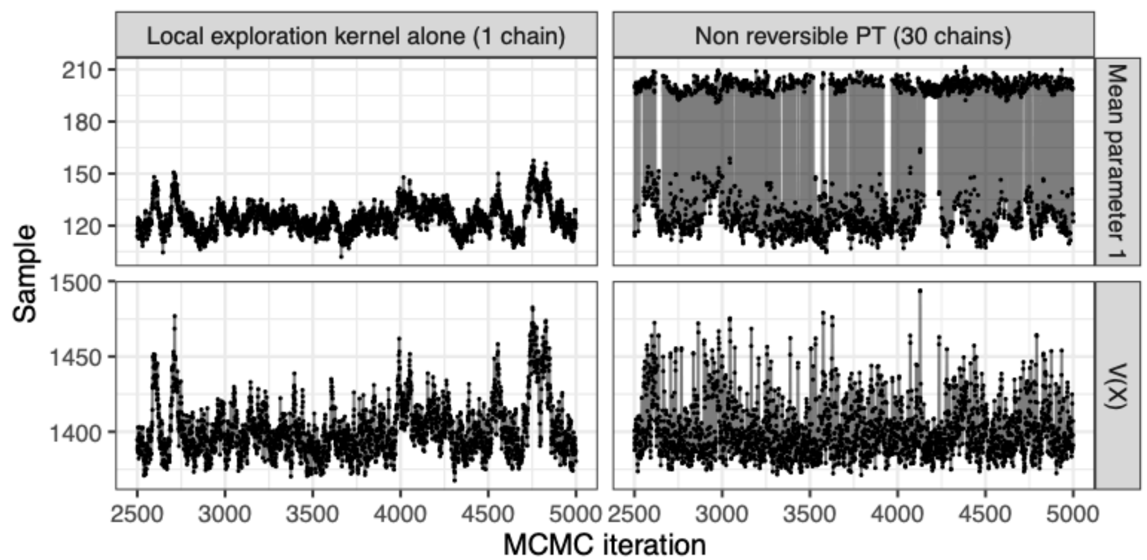


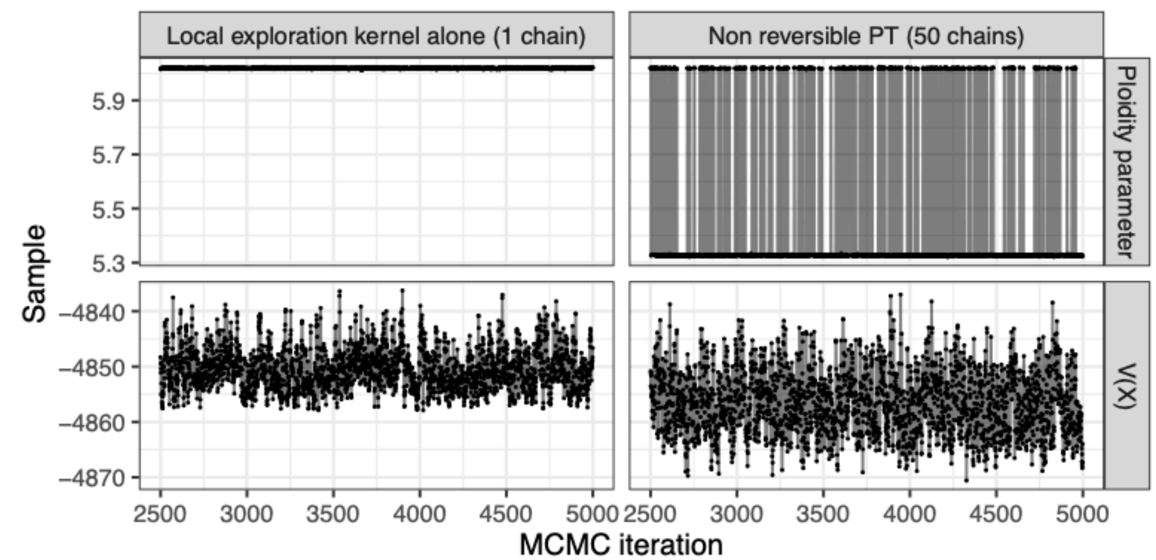
Ising model



Bayesian Mixture Model



Copy-number inference



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- **Theorem:** Let $r_n = s_{n,-}$ be the rejection probability for the swap between chains $n-1$ and n
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$$\mathbb{E}_{\mathrm{RPT}}[\Delta T] = 2N(N+1) + 2(N+1)\sum_{n=1}^{N}\frac{r_n}{1-r_n}$$

  - Hence the round trip rate satisfies:

$$\tau_{\mathrm{RPT}} = \frac{1}{2N + 2\sum_{n=1}^{N}\frac{r_n}{1-r_n}} \leq \frac{1}{2N}$$

  - Moreover as $N \to \infty$ we have $\tau_{\mathrm{RPT}} \to 0$ independent the schedule.
- **The first term** is expected round trip time for a simple random walk $2(N+1)N$
- **The second term** is the additional expected time due to rejections
- RPT is very sensitive to both the annealing scedule number of chains N
  - One high rection rate will make perfomance collapse
  - When N is small rejection dominate and N is large diffusivity dominates
- Difficult to tune or scale to modern hardware

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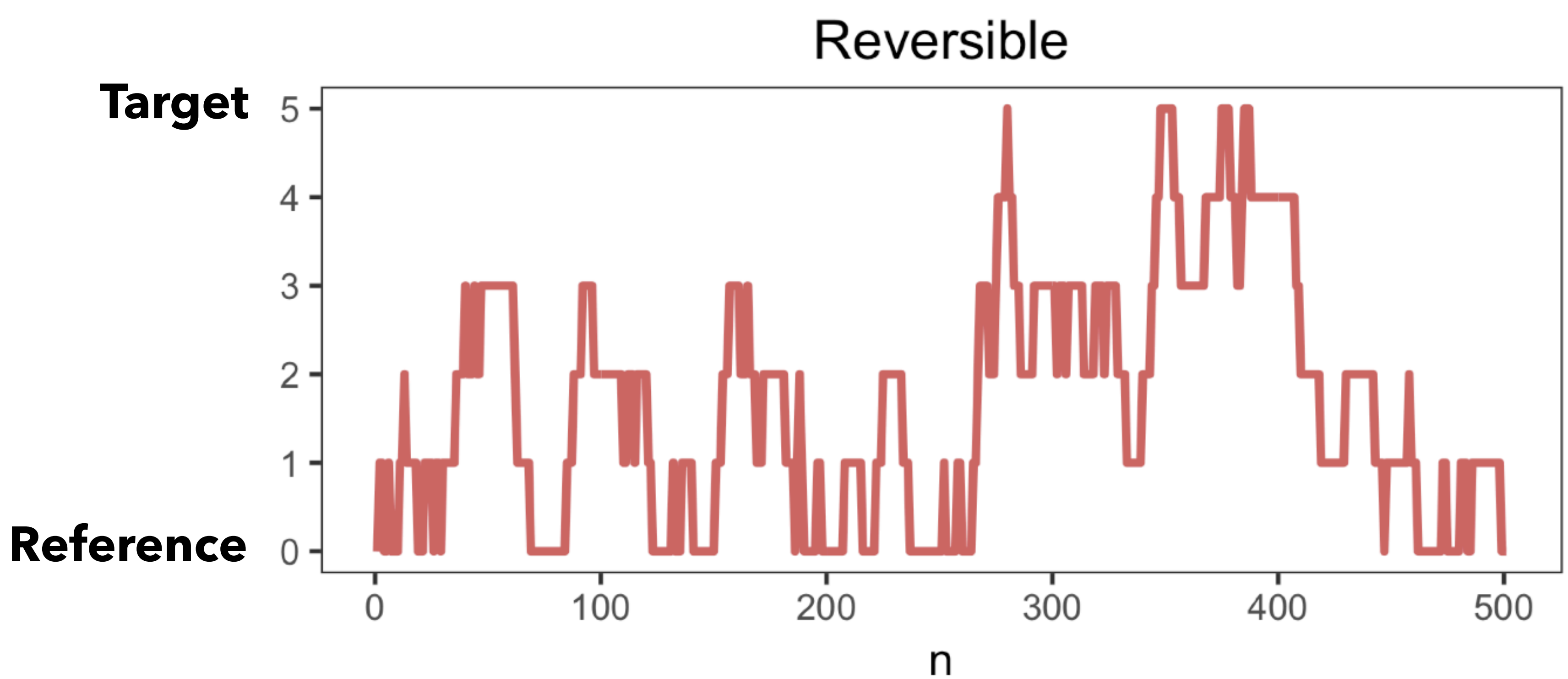
- Scales to GPUs' improves in performance increased paralellism (with marginal gains)

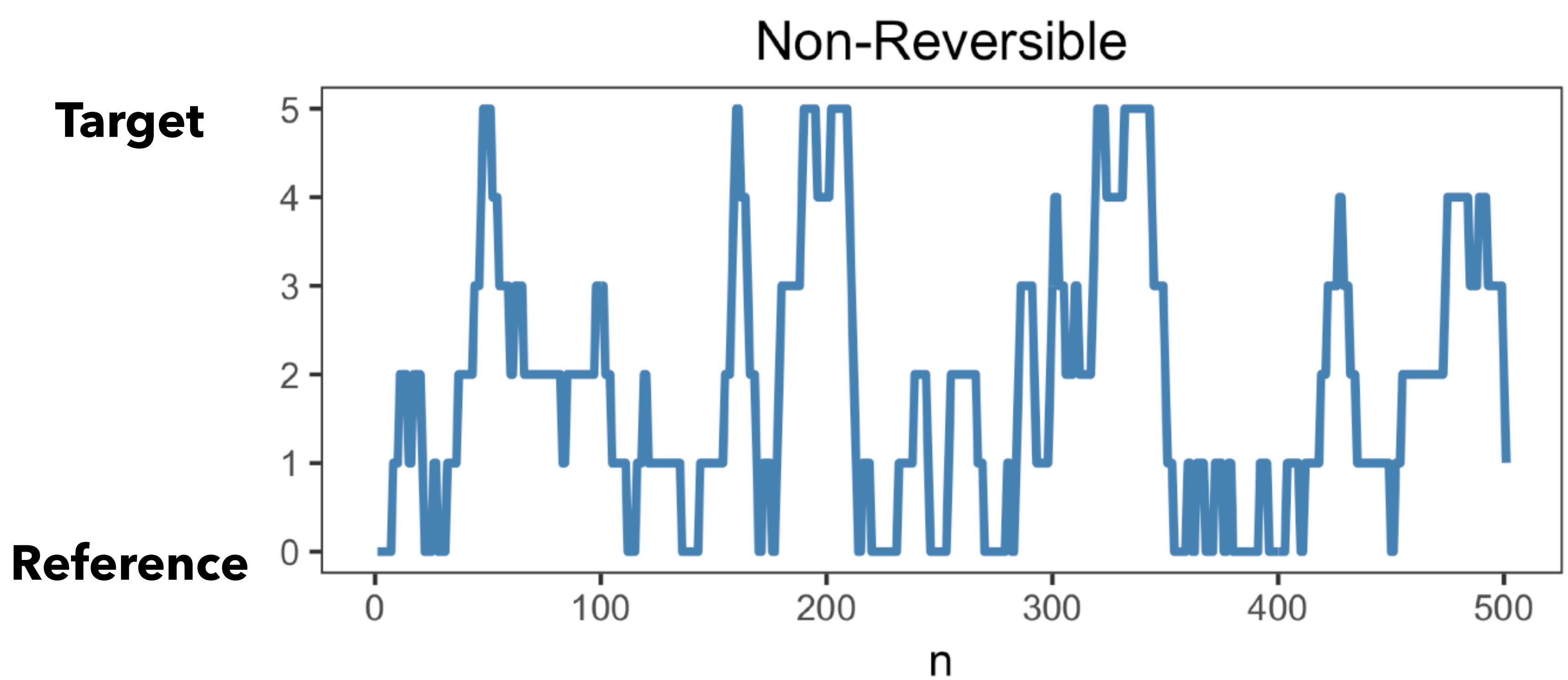
SAMPLE TRAJECTORIES

$$N = 5$$

 # Round Trips



1

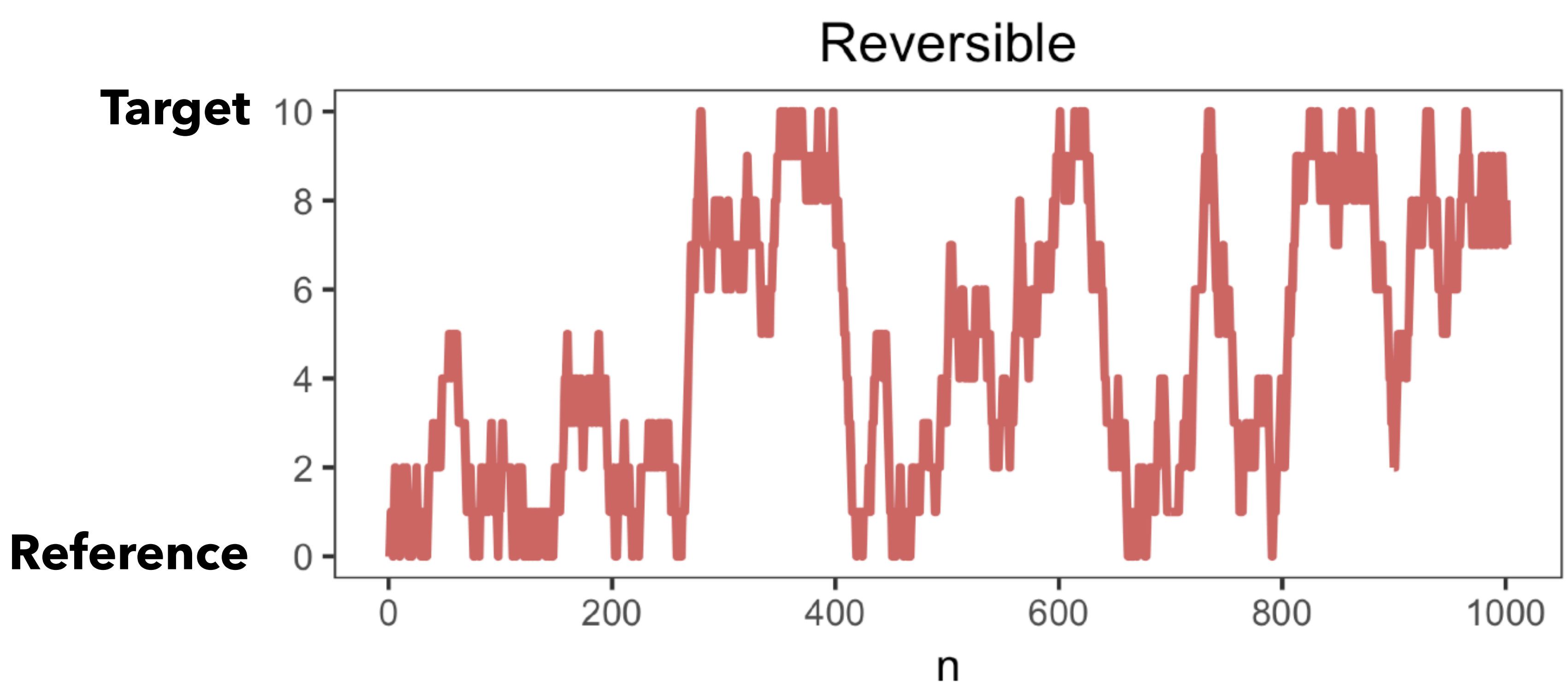


4

SAMPLE TRAJECTORIES

$$N = 10$$

 # Round Trips



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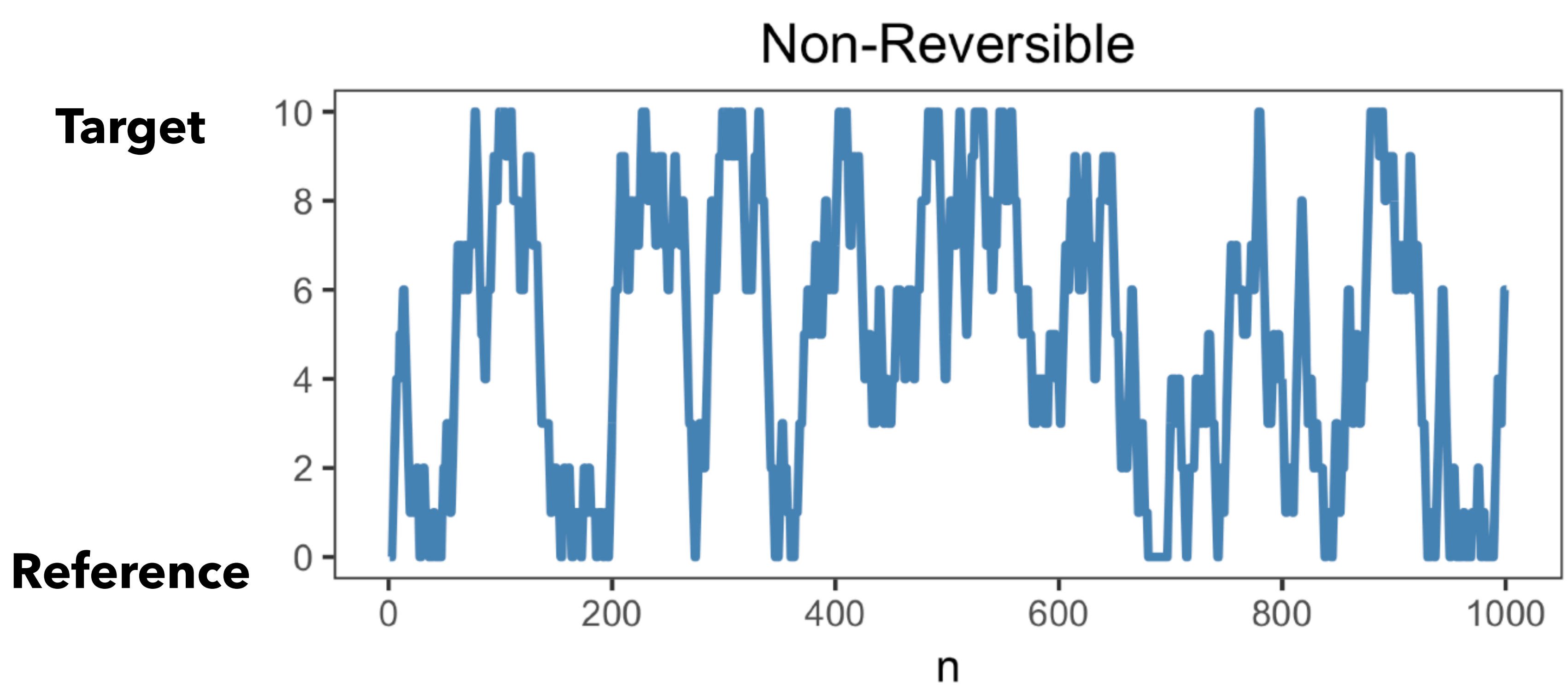
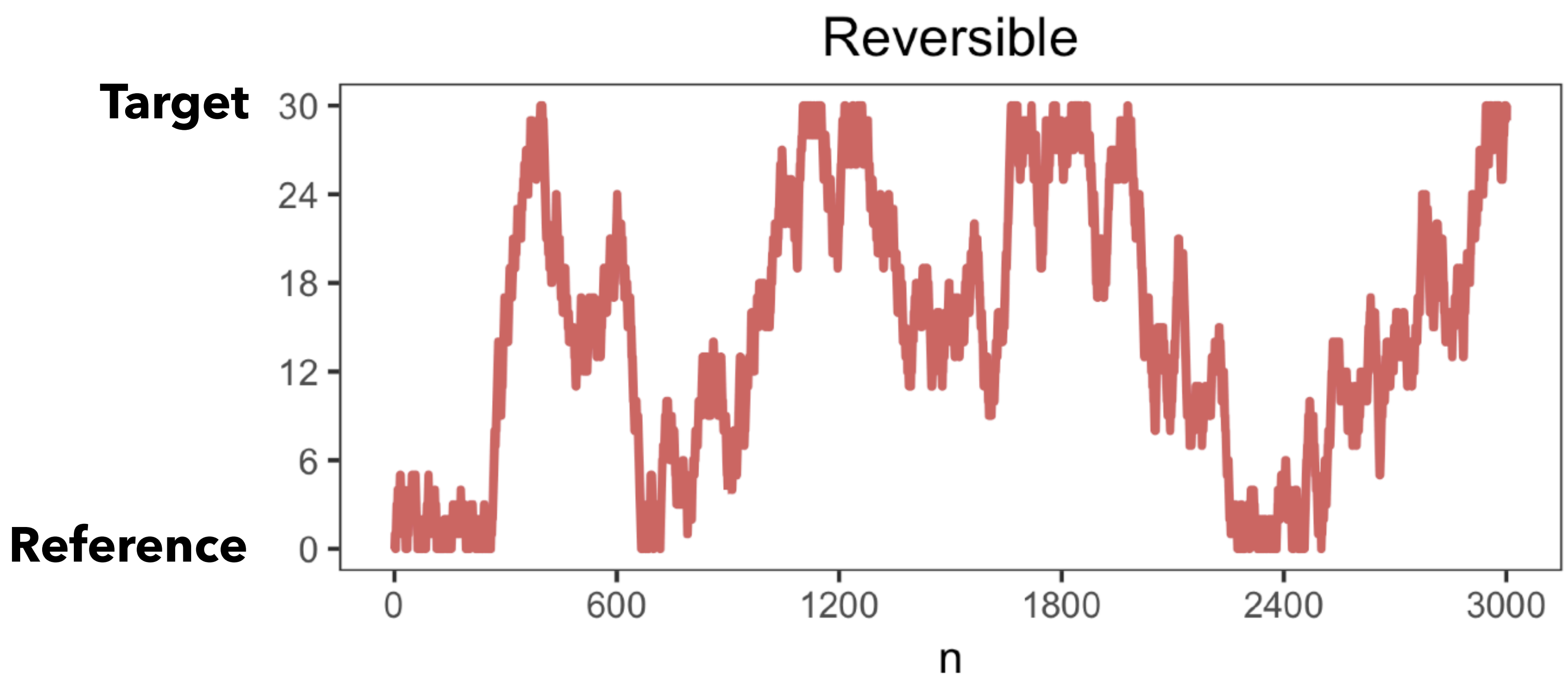


6

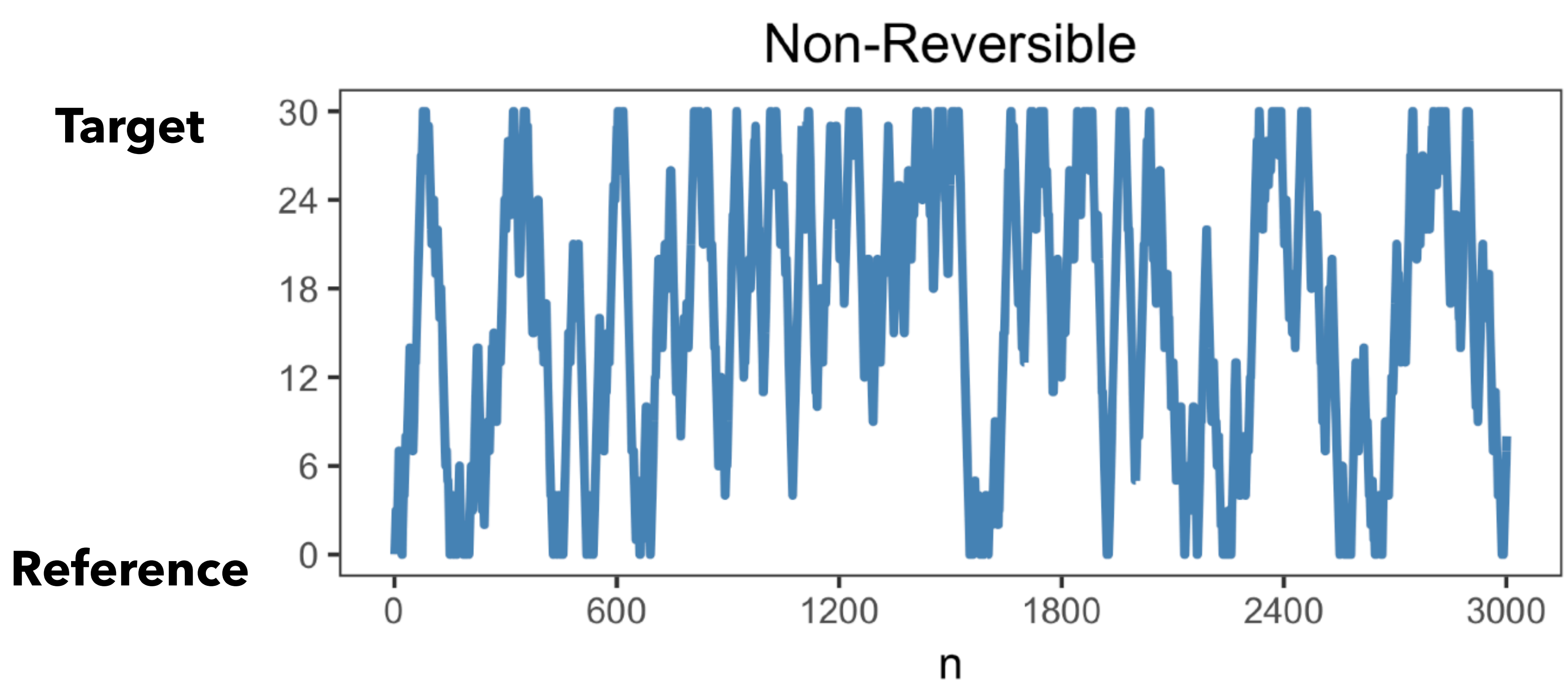
SAMPLE TRAJECTORIES

$$N = 30$$

 # Round Trips



$$2$$

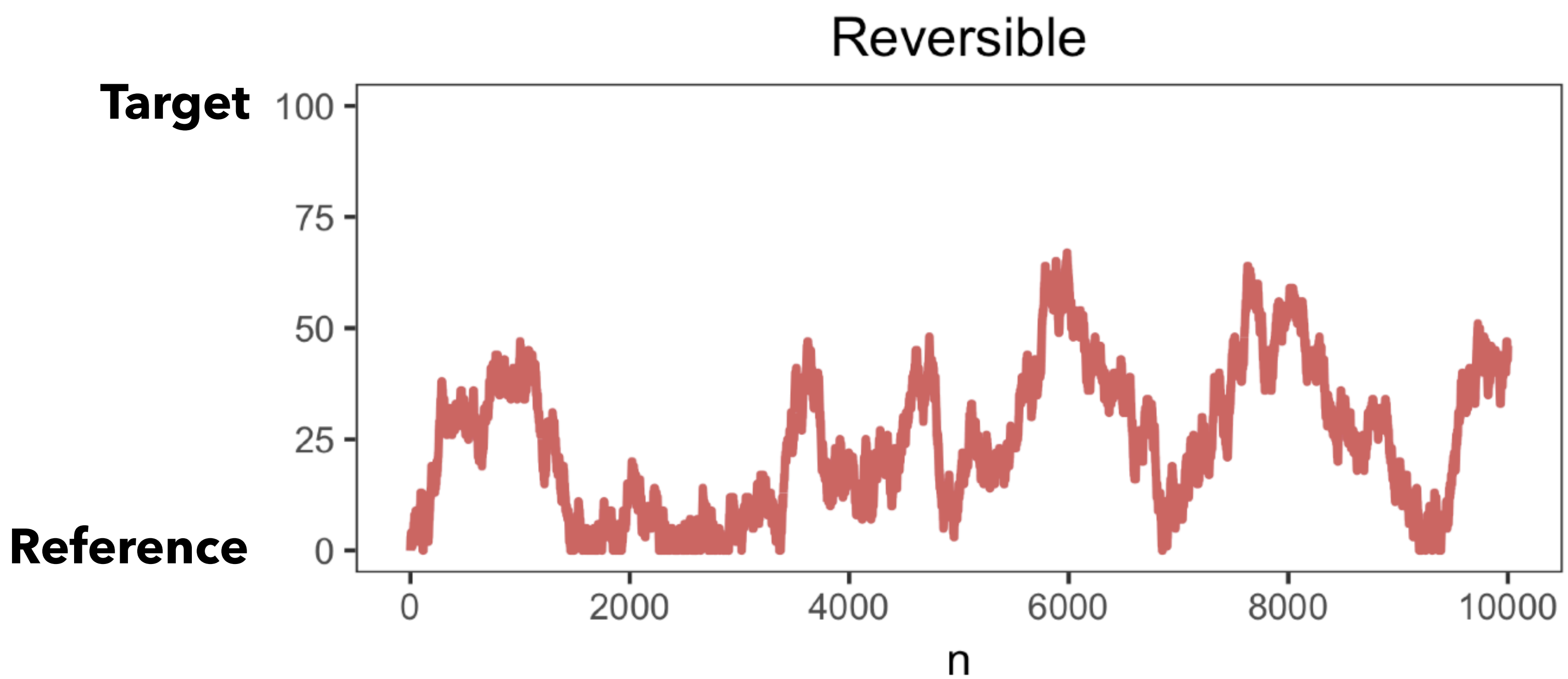


$$8$$

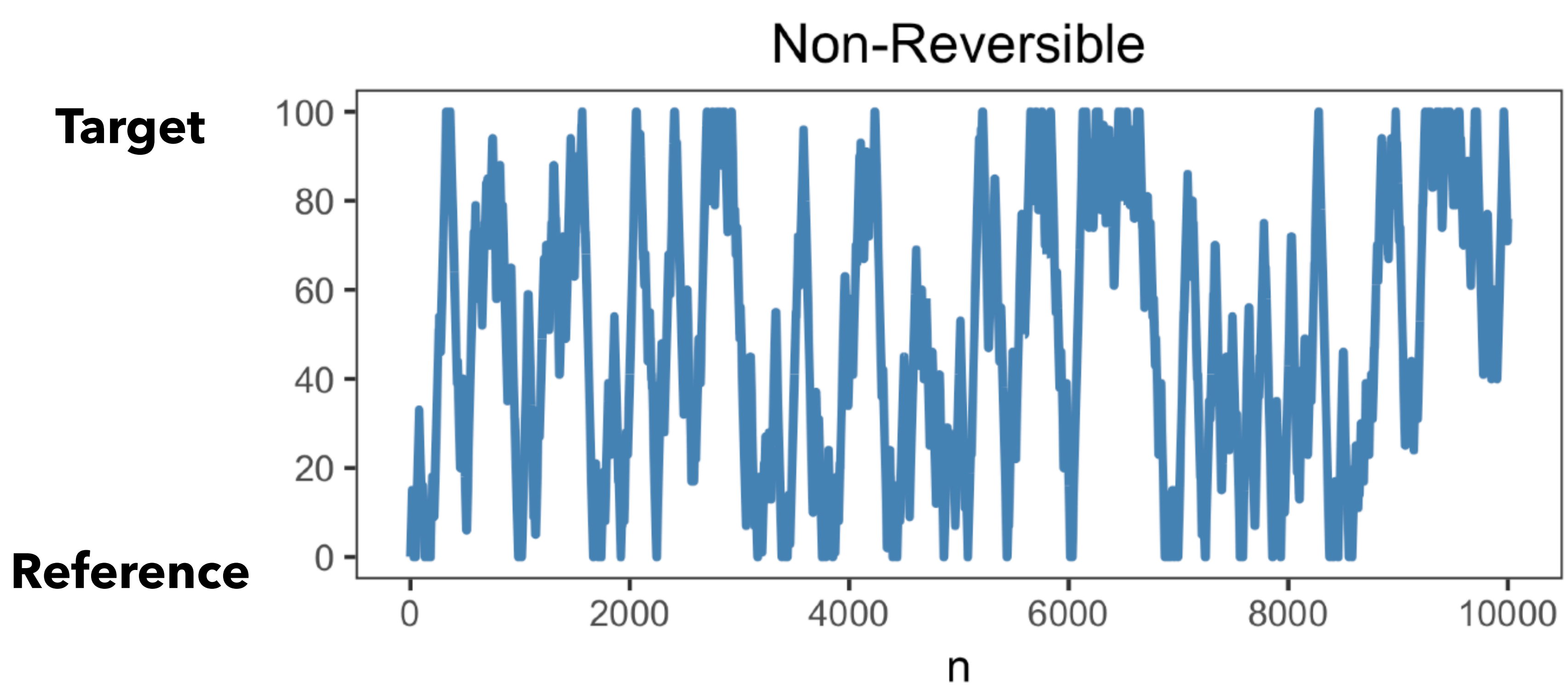
SAMPLE TRAJECTORIES

$$N = 100$$

 # Round Trips



$$0$$



$$9$$

SCALING LIMIT (REVERSIBLE)

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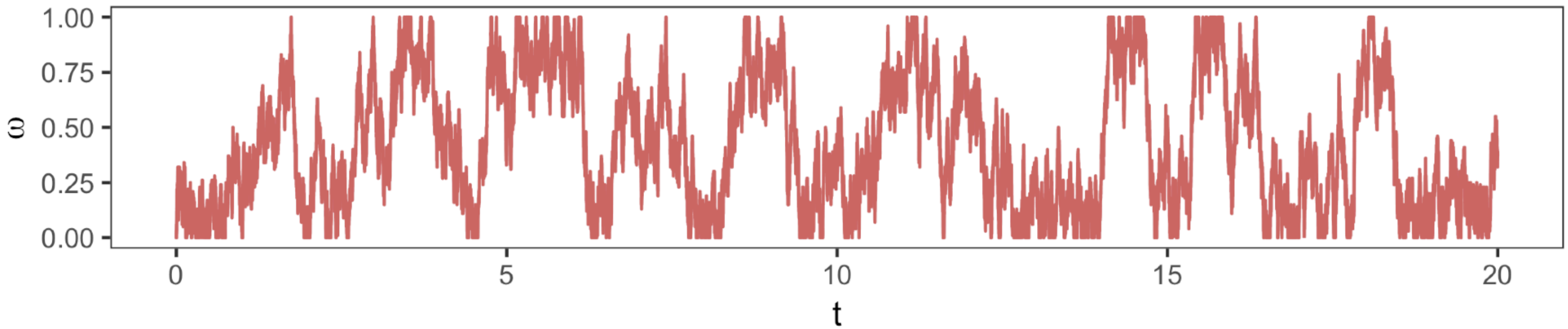

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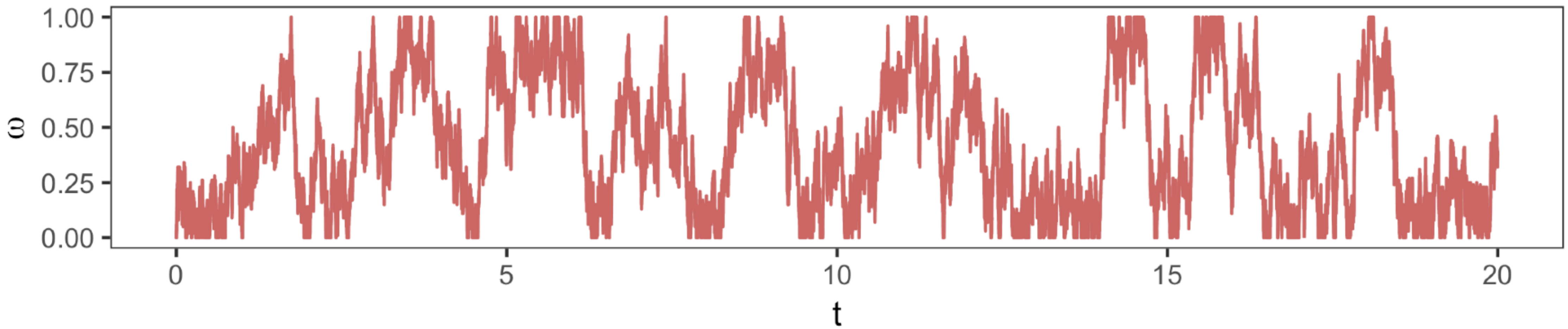

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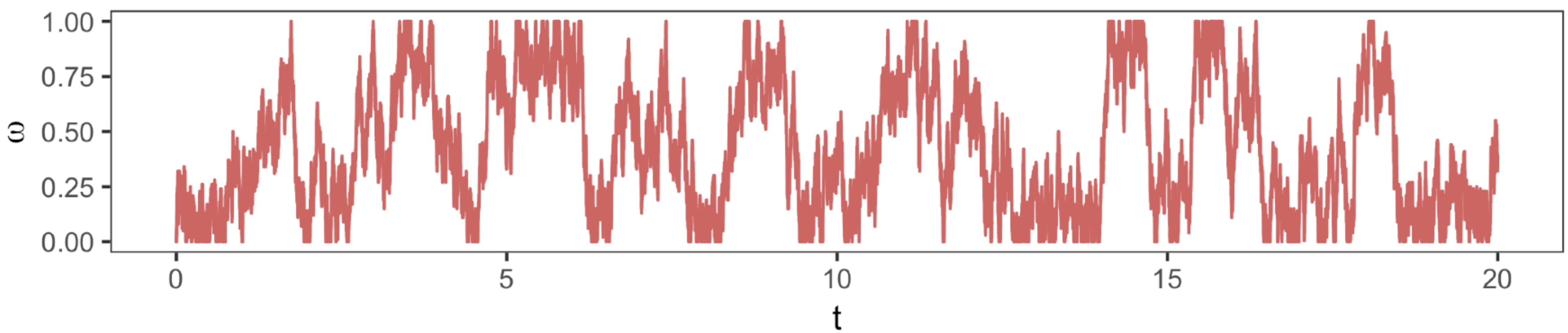

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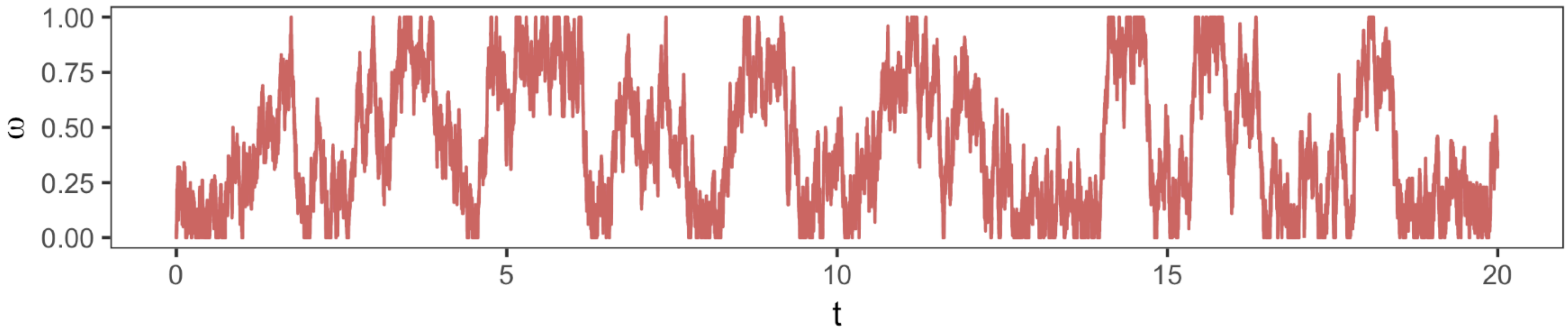

- As N grows large, noise from diffusivity dominates
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  - When N is large, modifying reference, target, or path leads to marginal gains

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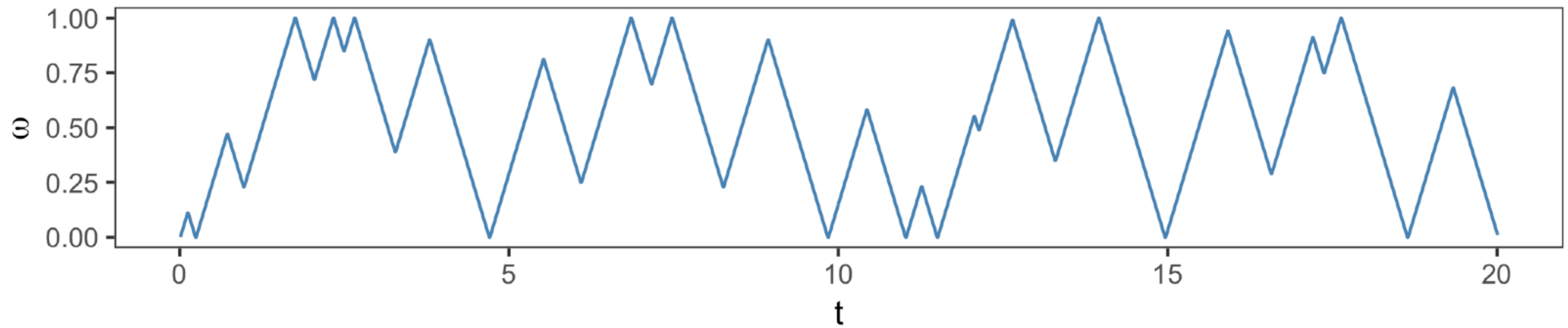

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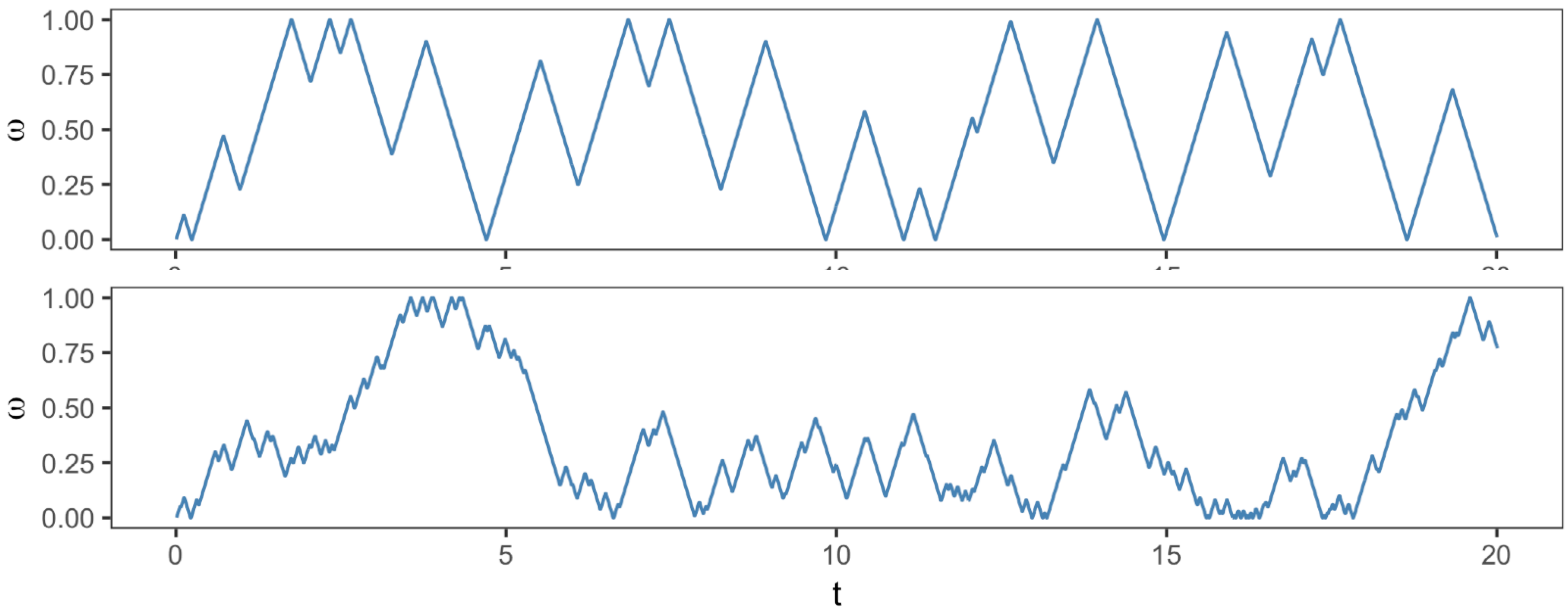

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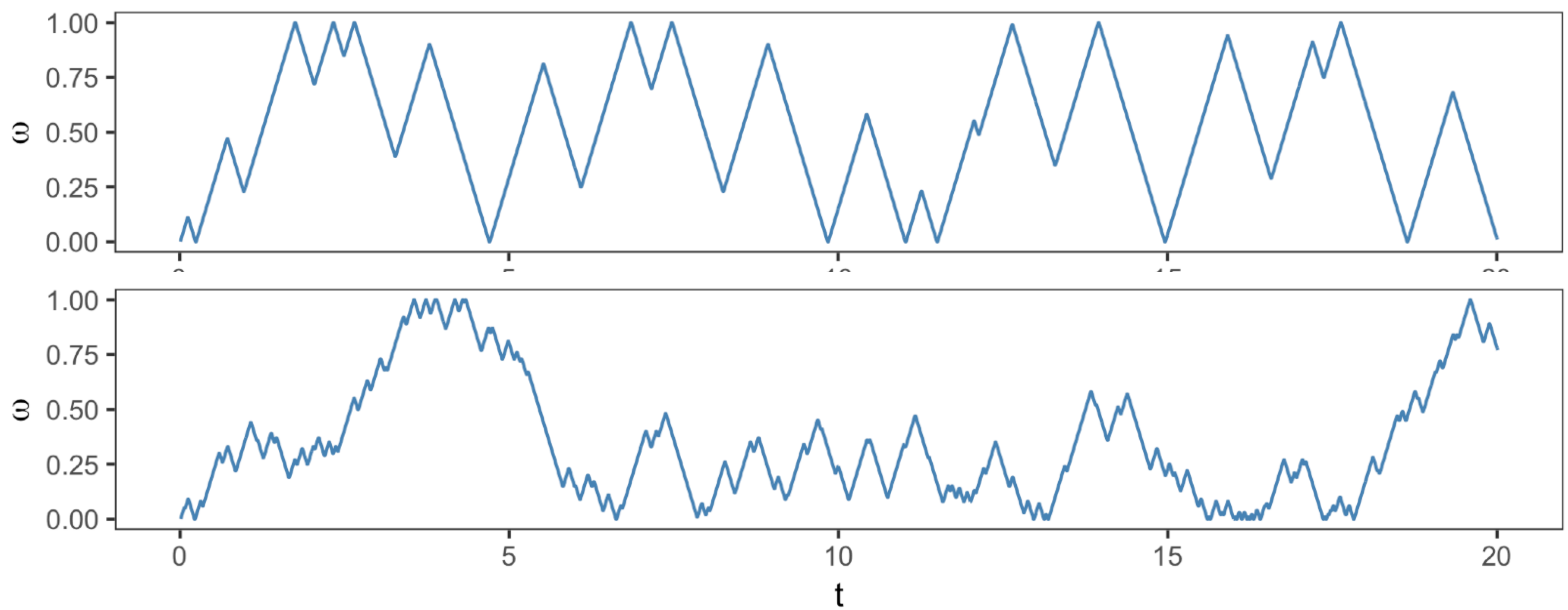

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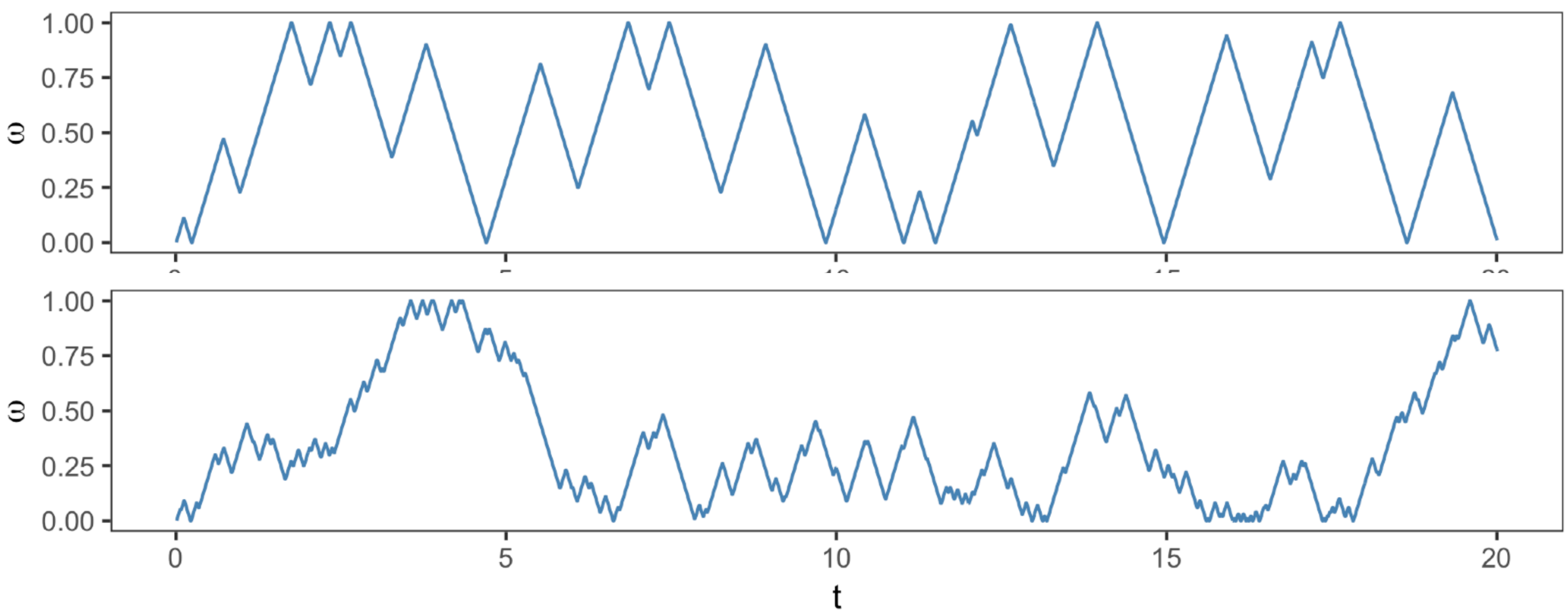

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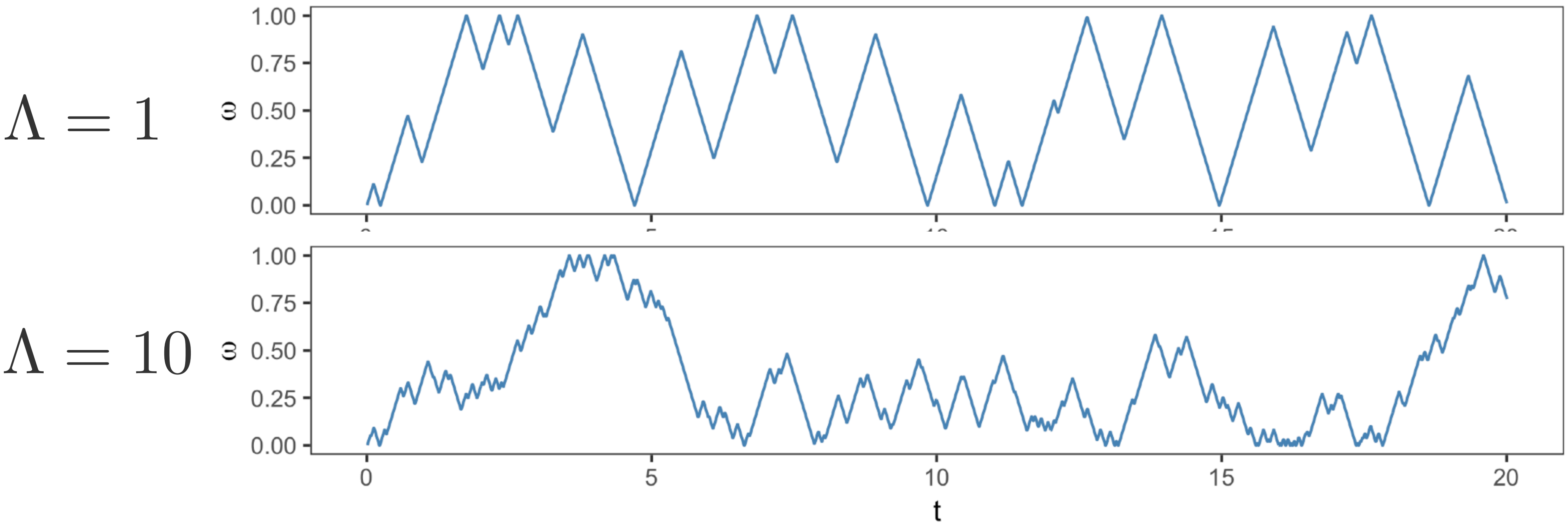

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- We can extend the rejection rate $r_n = r(\beta_{n-1}, \beta_n)$ for any β, β',

$$r(\beta, \beta') = 1 - \mathbb{E}\left[1 \wedge \frac{\pi_\beta(X')\pi_{\beta'}(X)}{\pi_\beta(X)\pi_{\beta'}(X')}\right], \qquad (X, X') \sim \pi_\beta \otimes \pi_{\beta'}$$

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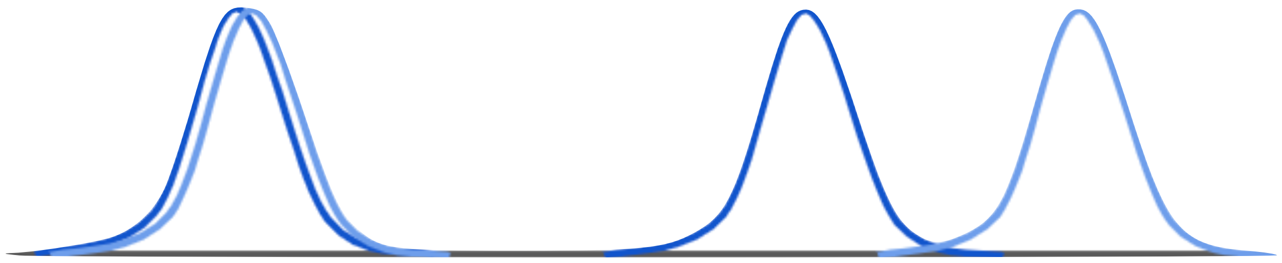

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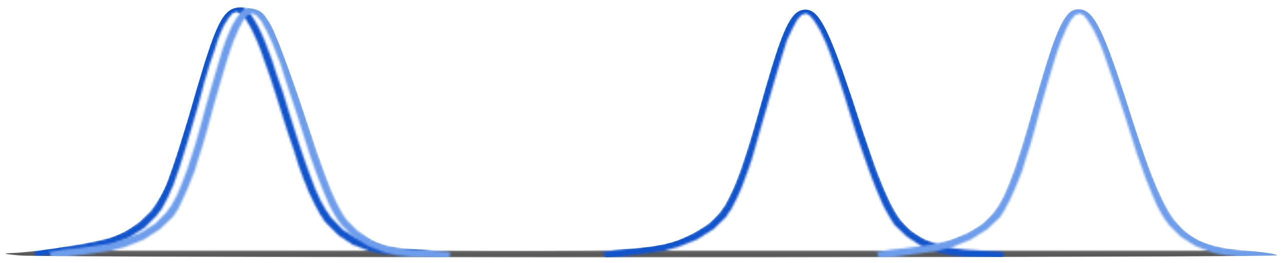

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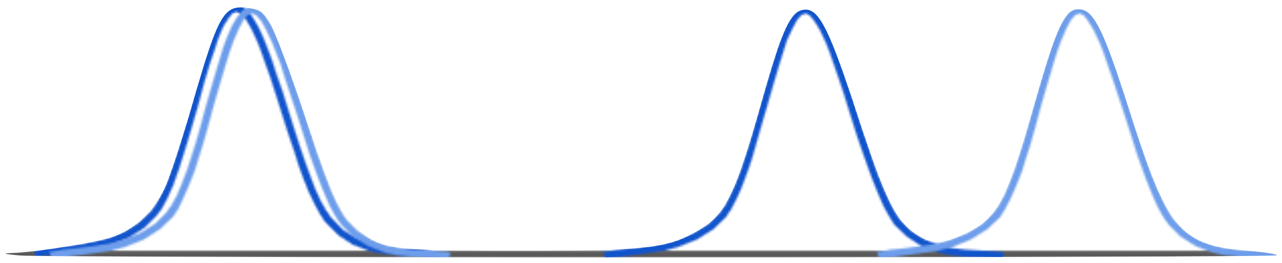

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$$\lambda(\beta) = \frac{1}{2}\mathbb{E}[\,|\,\dot{\pi}_\beta(X') - \dot{\pi}_\beta(X)\,|\,], \qquad (X, X') \sim \pi_\beta \otimes \pi_\beta$$

  - Defines the speed of the path with respect to the rejection!

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- The global barrier controlls the asymptotic efficiency of PT

$$\lim_{N\to\infty} \tau_{\mathrm{NRPT}} = \frac{1}{2 + 2\Lambda}$$

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- Recall that the round trip rate is a symmetric function of the rejection rates $r_n = r(\beta_{n-1}, \beta_n)$

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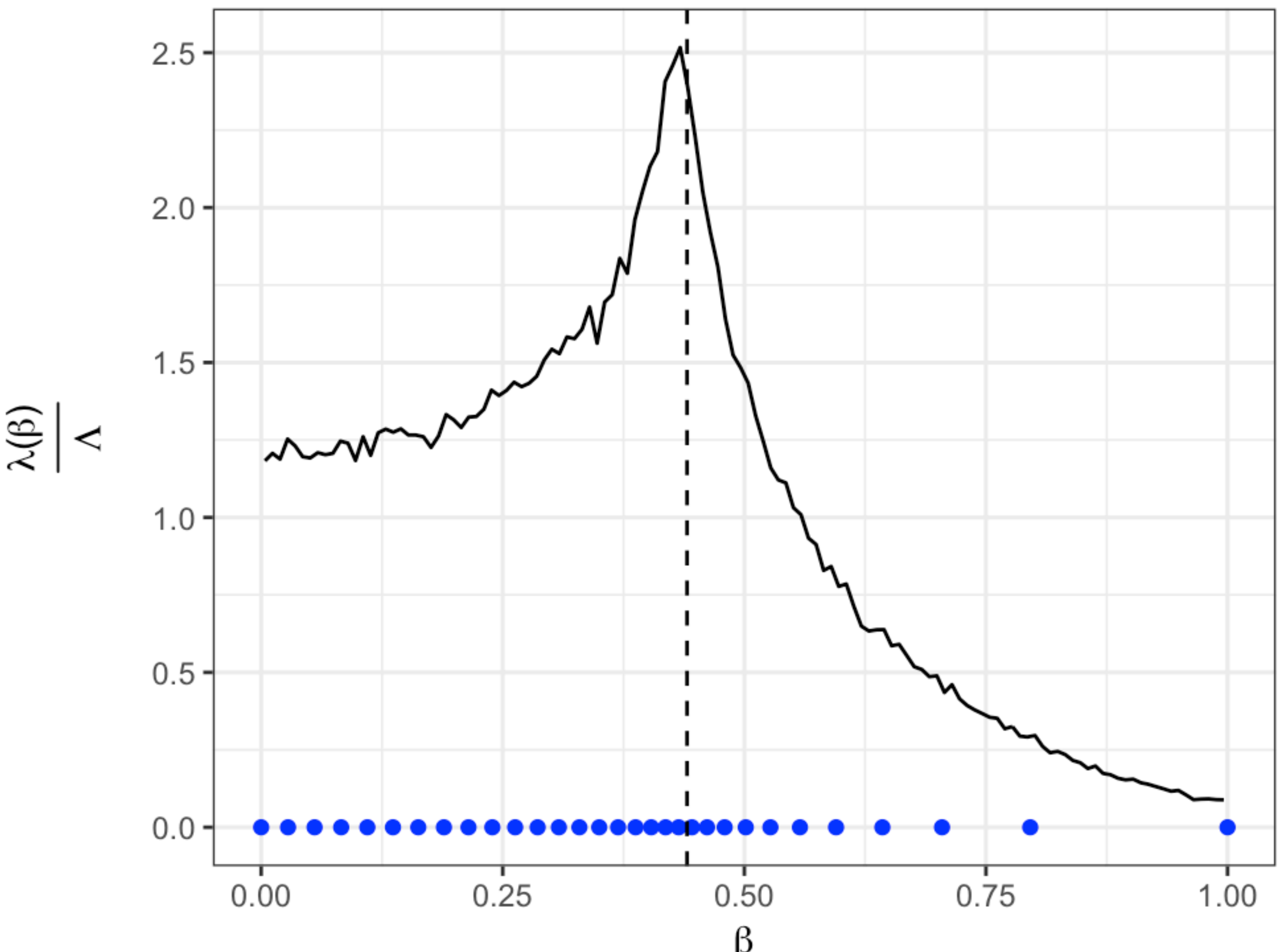
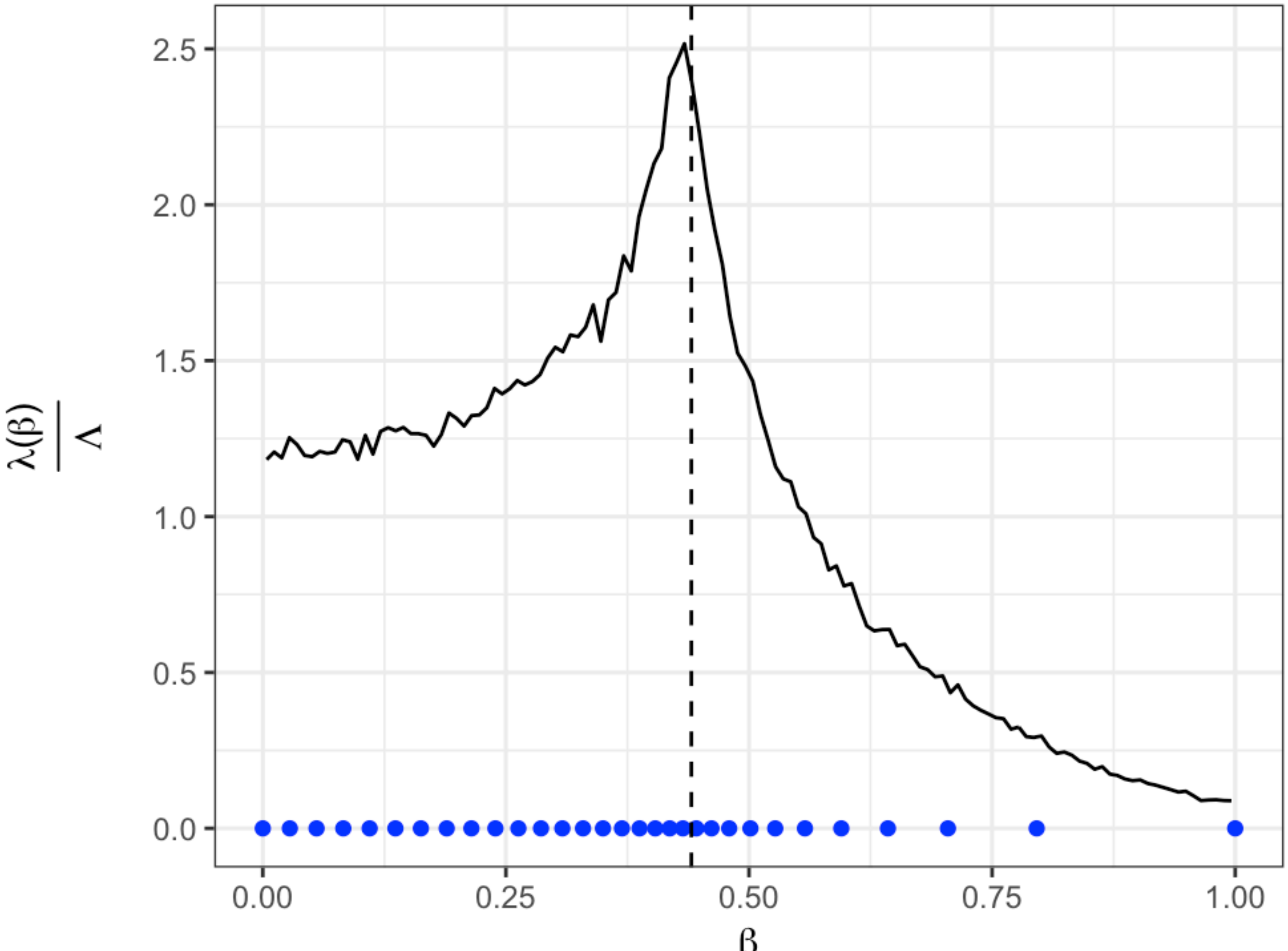
  - Optimal schedule accumulates constant length traverses path with constant speed.

EXAMPLE: ISING MODEL (D = 400)



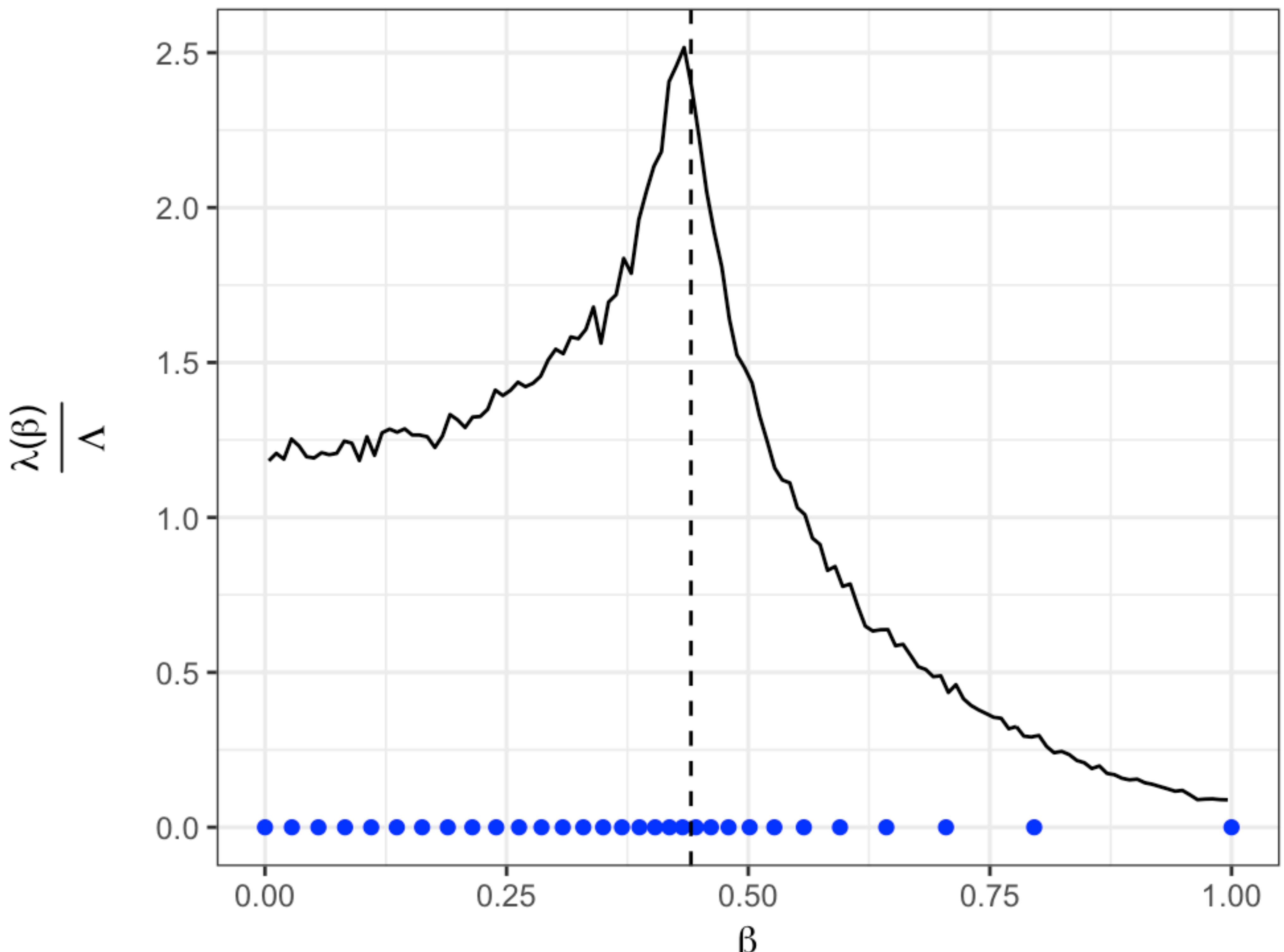
EXAMPLE: ISING MODEL (D = 400)

- An optimal schedule accumulates a constant-length between parameters



EXAMPLE: ISING MODEL (D = 400)

- An optimal schedule accumulates a constant-length between parameters
- Spaces annealing schedule with a density proportional to $\lambda(\beta)$



ALGORITHM: UPDATE SCHEDULE

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$$\hat{r}(\beta_{n-1}, \beta_n) \leftarrow \texttt{PT}(\mathcal{B}_N)$$

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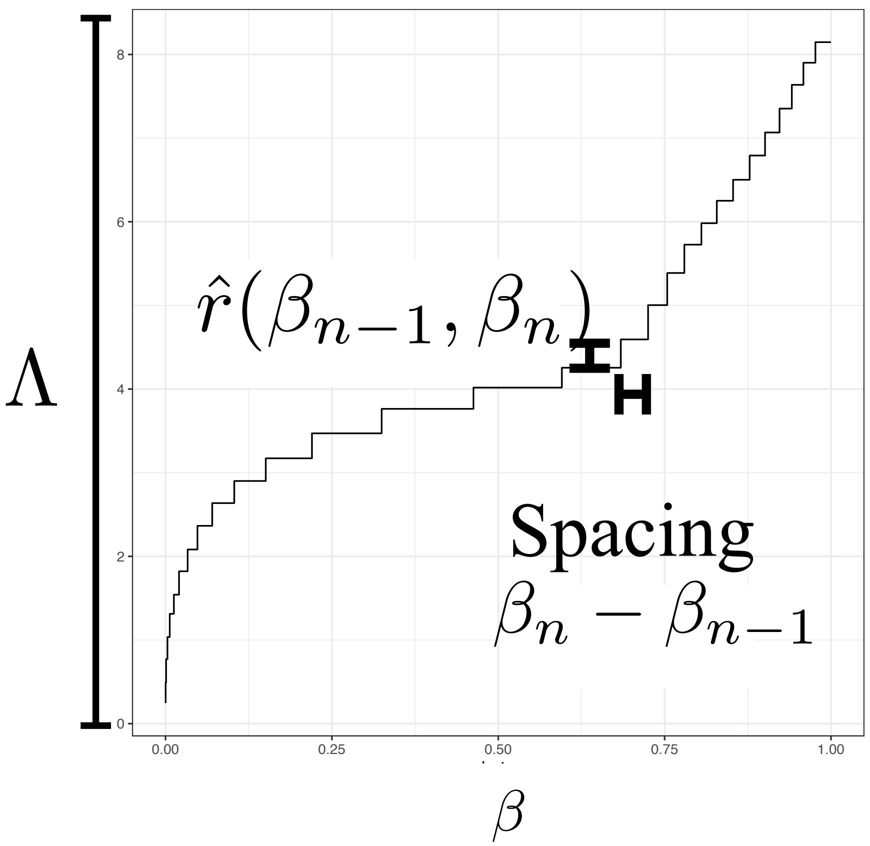

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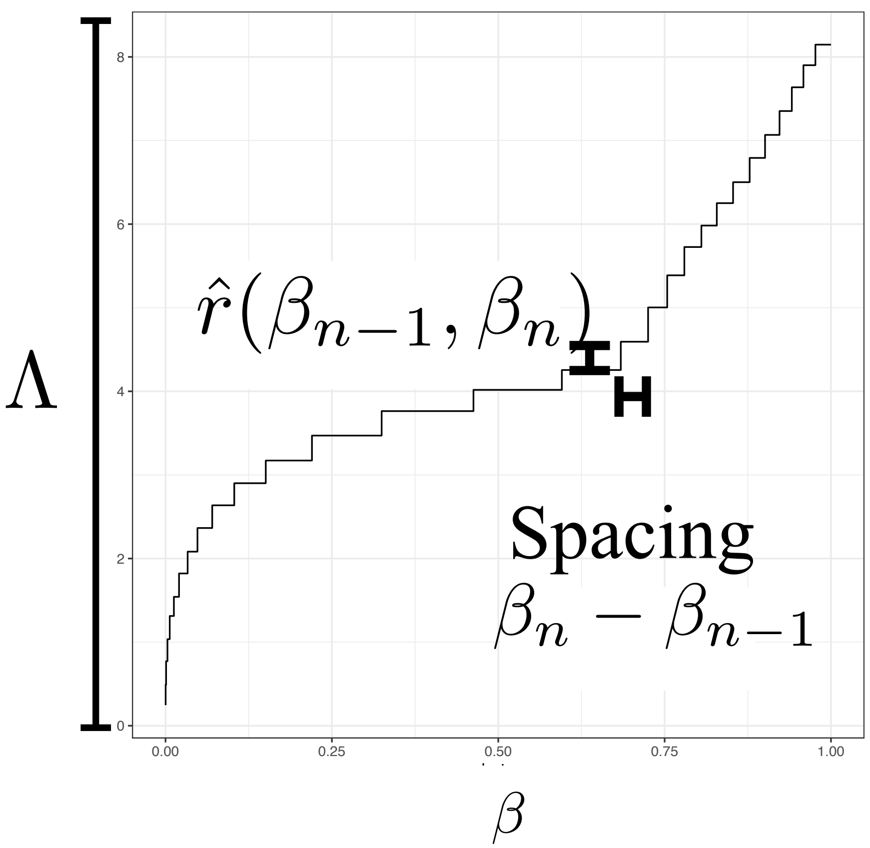

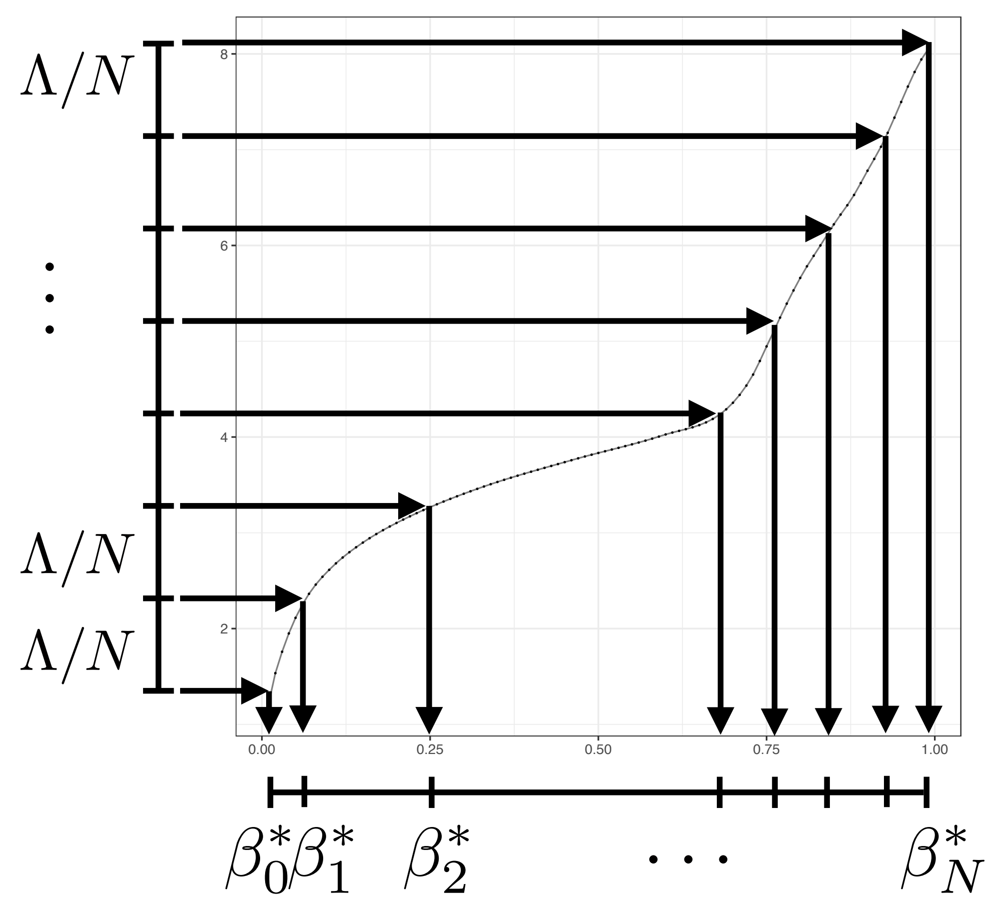

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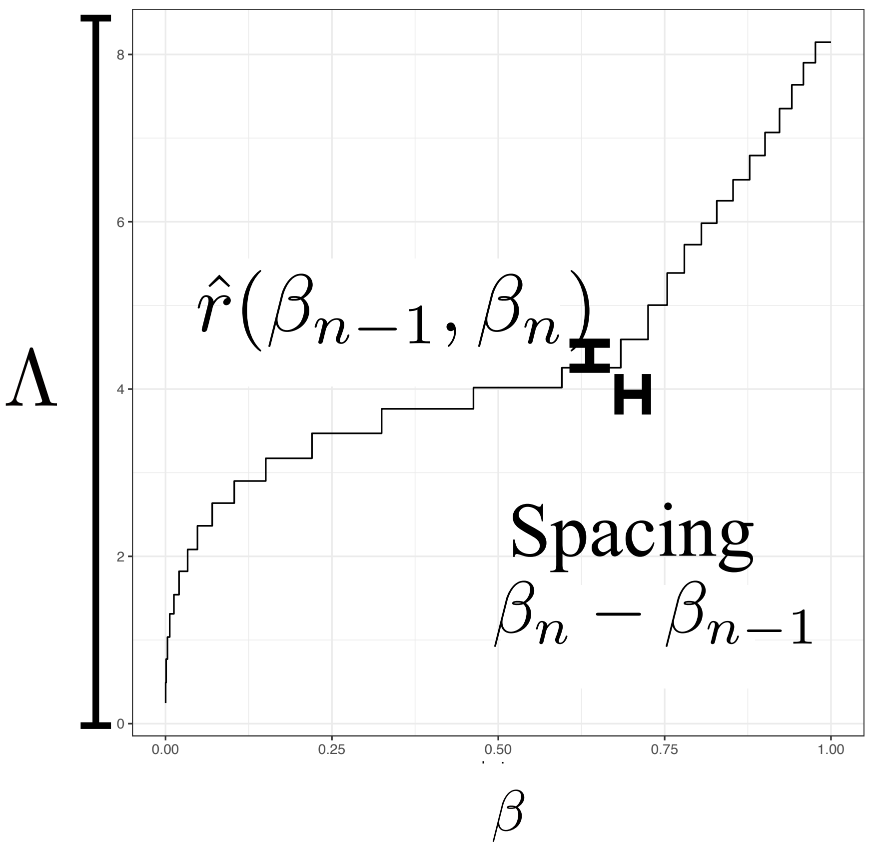
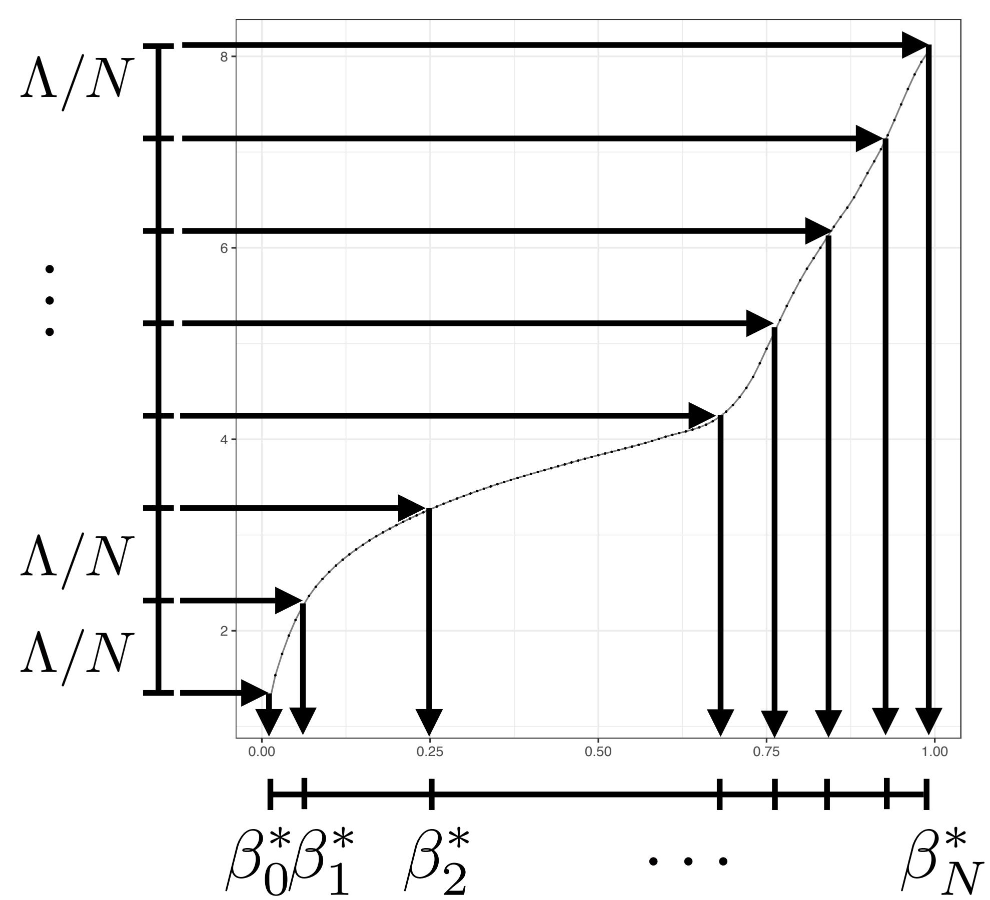
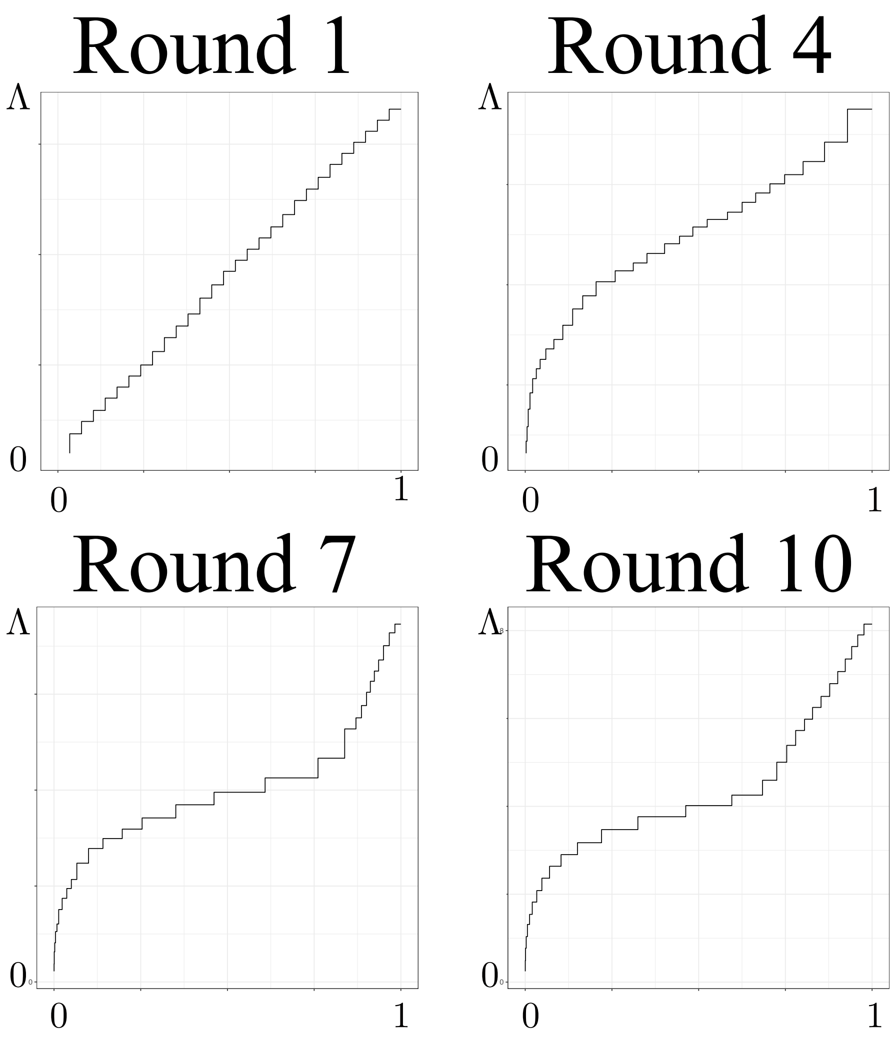
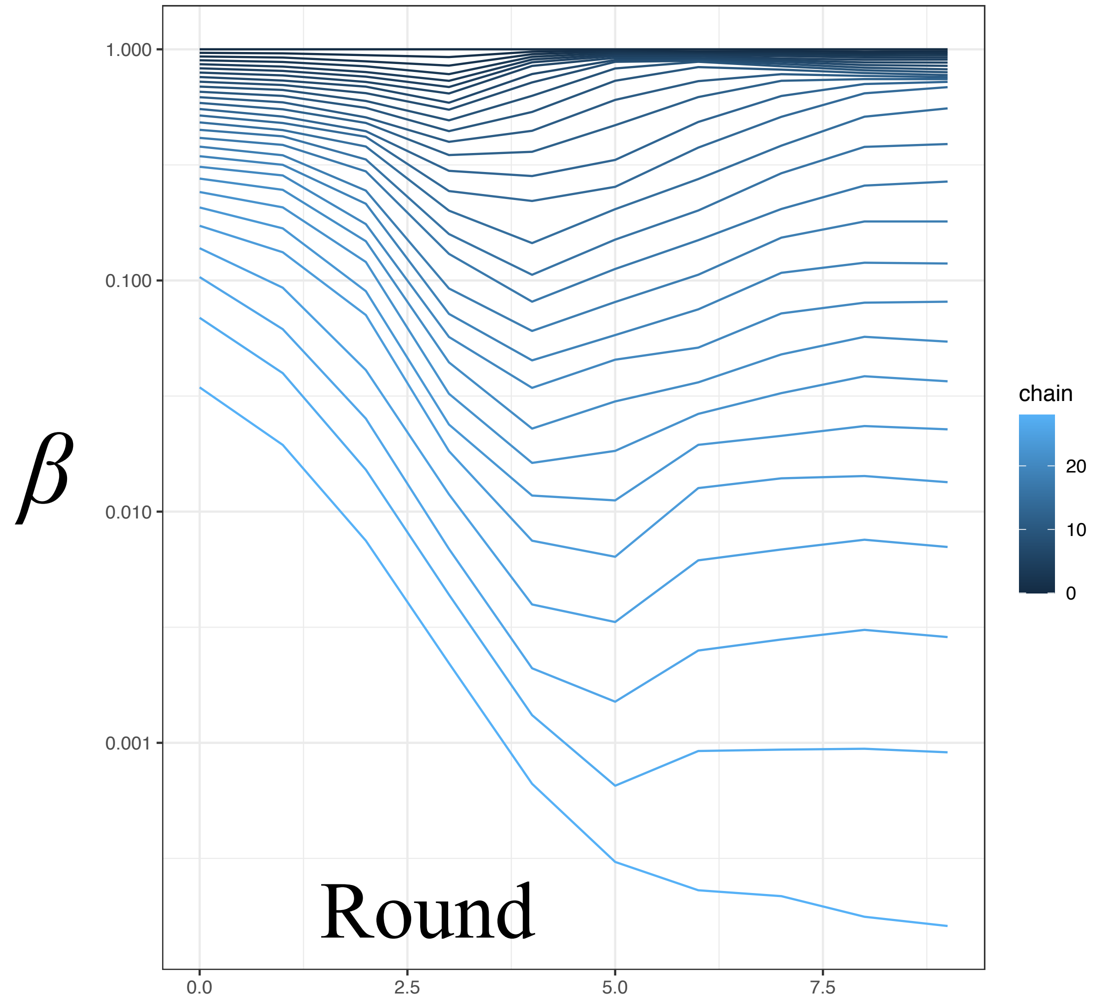


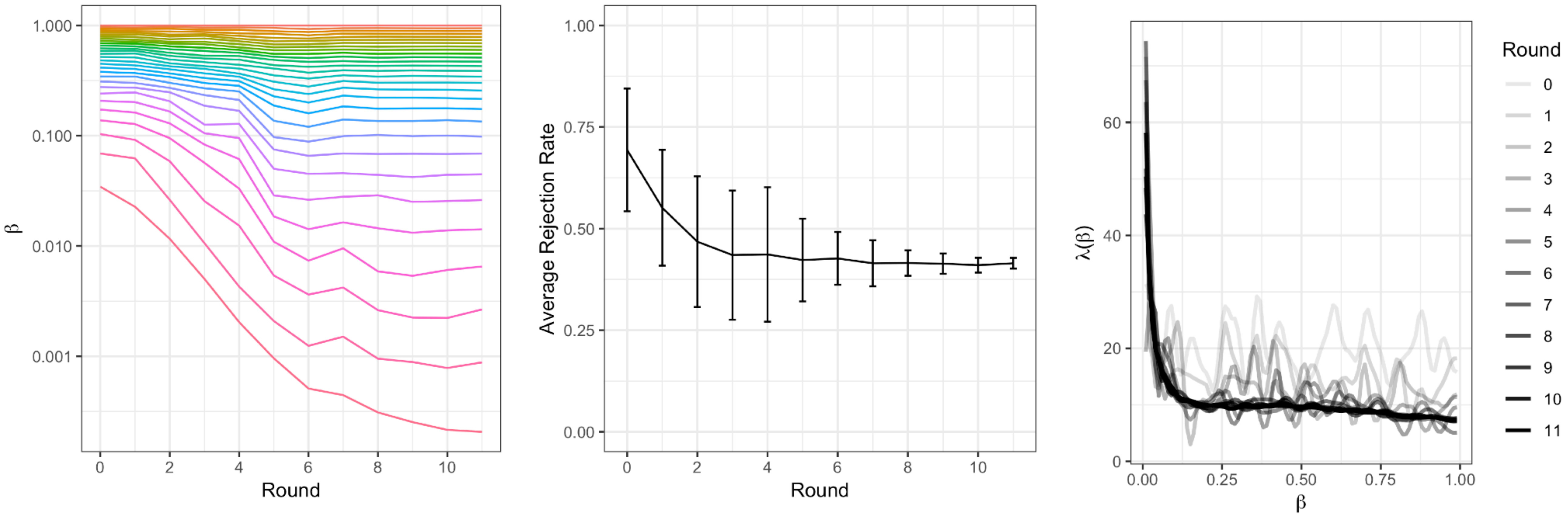


EXAMPLE: HIERARCHICAL BAYESIAN MODEL (D = 369)





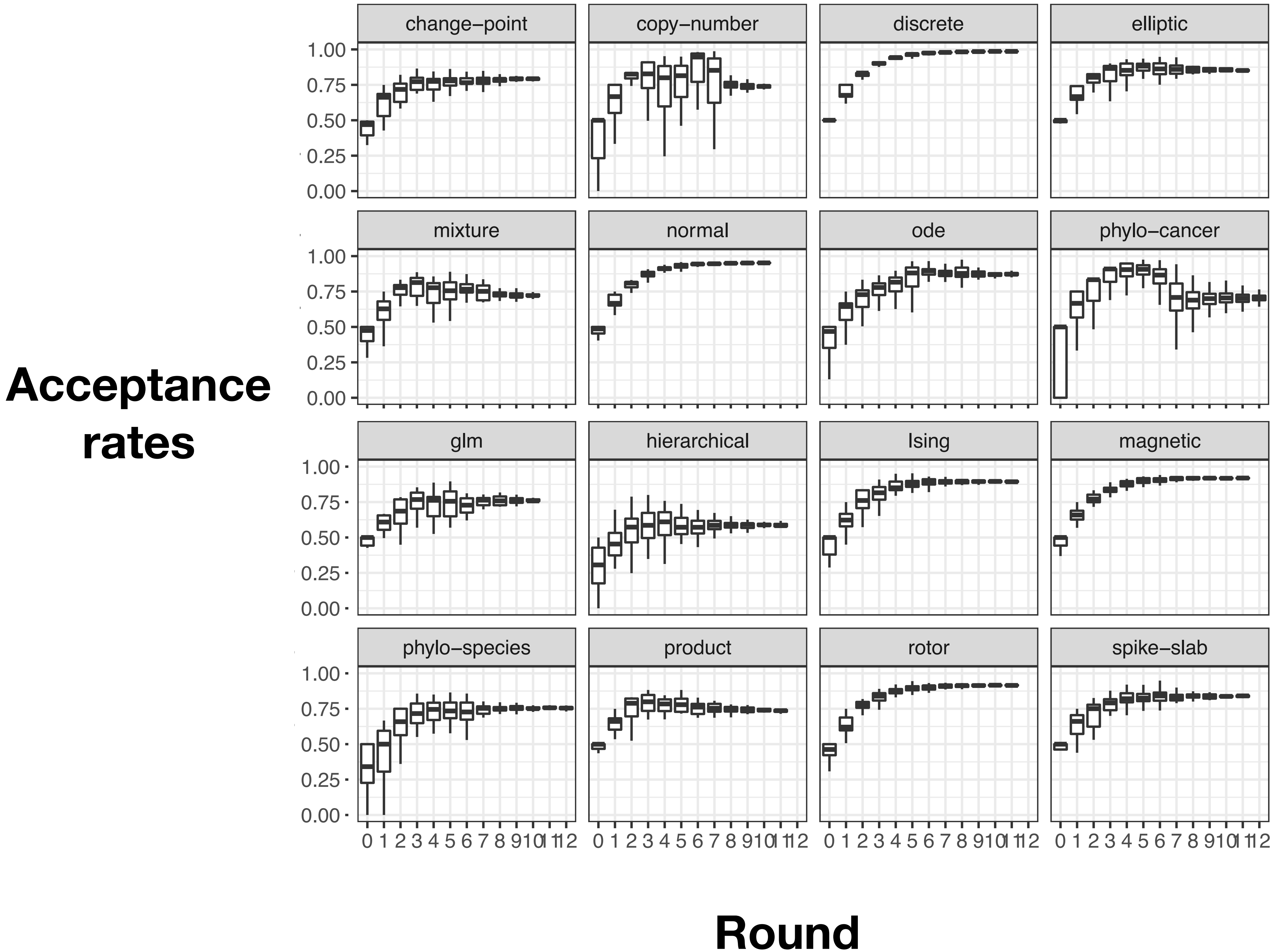
EXAMPLE: HIERARCHICAL BAYESIAN MODEL (D = 369)



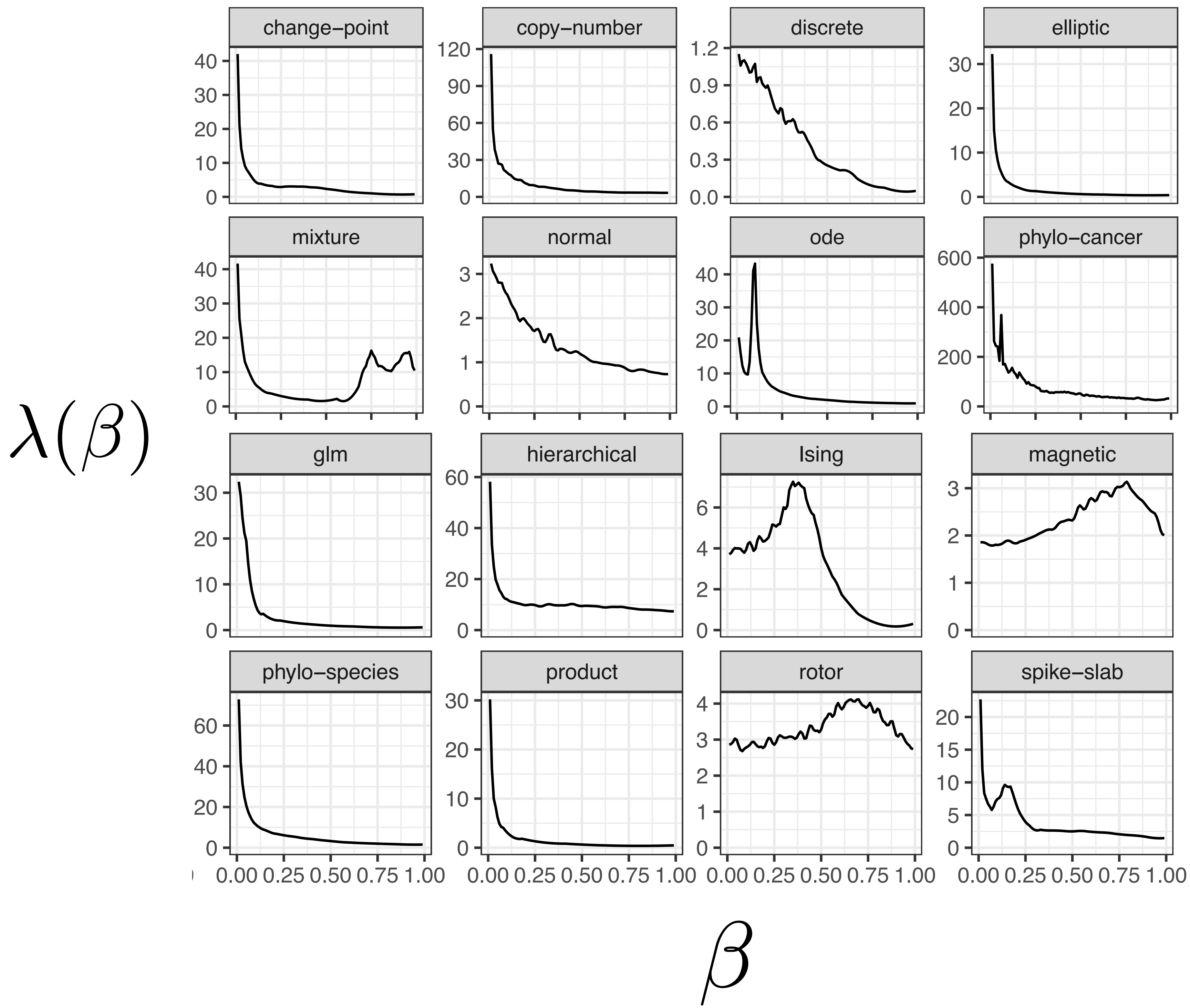
NRPT IN PRACTICE

Model (and dataset when applicable)	n	d	$\hat{\Lambda}$	N
Change point detection (text message data, [Davidson-Pilon, 2015])	74	3	4.0	20
Copy number inference (whole genome ovarian cancer data, Section 7.5)	6 206	30	13.0	50
Discrete multimodal distribution (Appendix I.4.3)	N/A	3	0.4	30
Weakly identifiable *elliptic* curve (Appendix I.4)	N/A	2	4.4	30
General Linear Model (*GLM*) (Challenger O-ring dataset, [Dalal et al., 1989])	23	2	3.3	15
Bayesian *hierarchical* model (historical rocket failure data, Appendix I.4)	5 667	369	12.0	30
Ising model (Appendix I.4.2)	N/A	25	3.1	30
Ising model with *magnetic* field (Appendix I.4.2)	N/A	25	2.3	30
Bayesian *mixture* model (Section I.4.4)	300	305	8.2	30
Bayesian *mixture* model (subset of 150 datapoints)	150	155	5.5	20
Isotropic *normal* distribution	N/A	5	1.4	30
ODE parameters (mRNA data, [Leonhardt et al., 2014])	52	5	6.4	50
Phylogenetic inference (single cell breast cancer data, [Dorri et al., 2020])	192 763	192 765	88	300
Phylogenetic species tree inference (mtDNA, [Hayasaka et al., 1988])	249	10 395	7.1	30
Unidentifiable *product* parameterization (Appendix I.4)	100 000	2	3.7	15
Rotor (XY) model, [Hsieh et al., 2013]	N/A	25	3.3	40
Spike-and-slab classification (RMS Titanic passengers data [Hind, 2019])	200	19	4.7	30

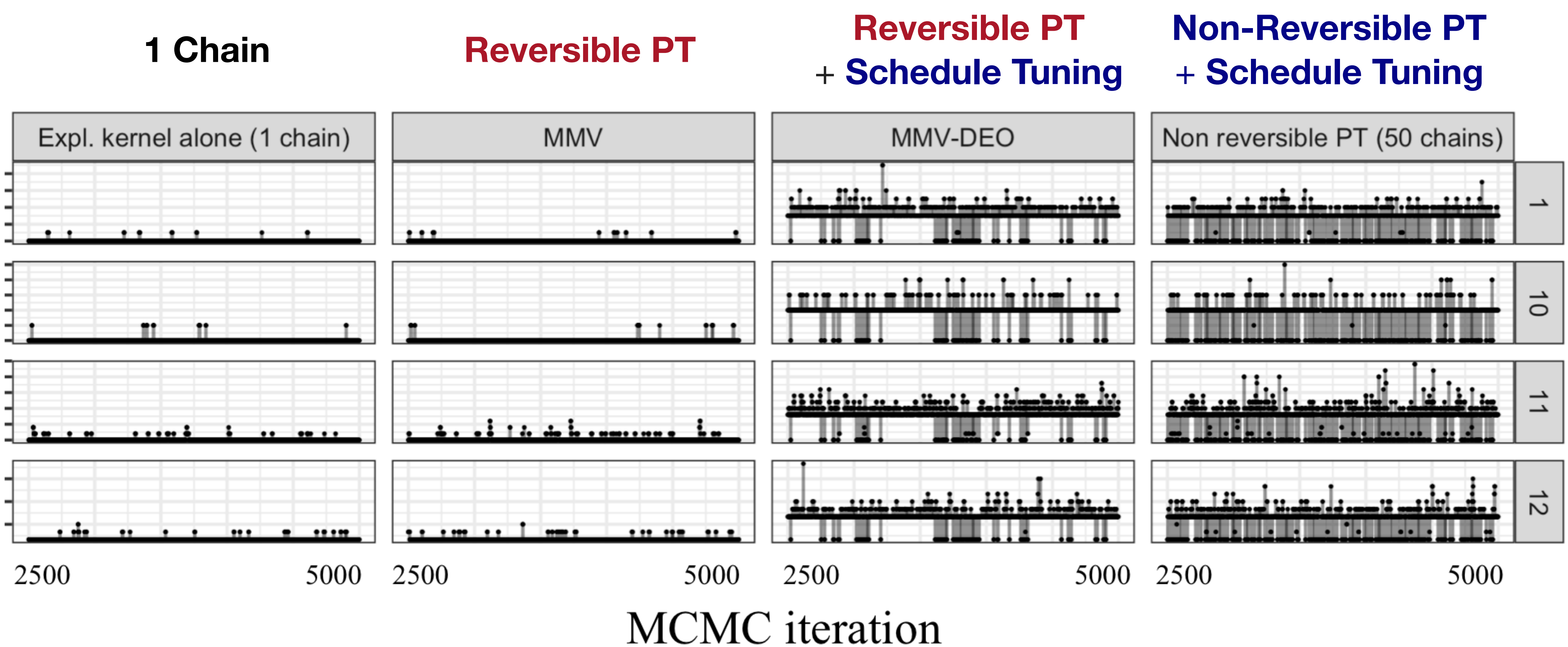
NRPT IN PRACTICE: CONVERGENCE



NRPT IN PRACTICE: LOCAL COMMUNICATION BARRIER



EXAMPLE: COPY NUMBER INFERENCE (D = 30)



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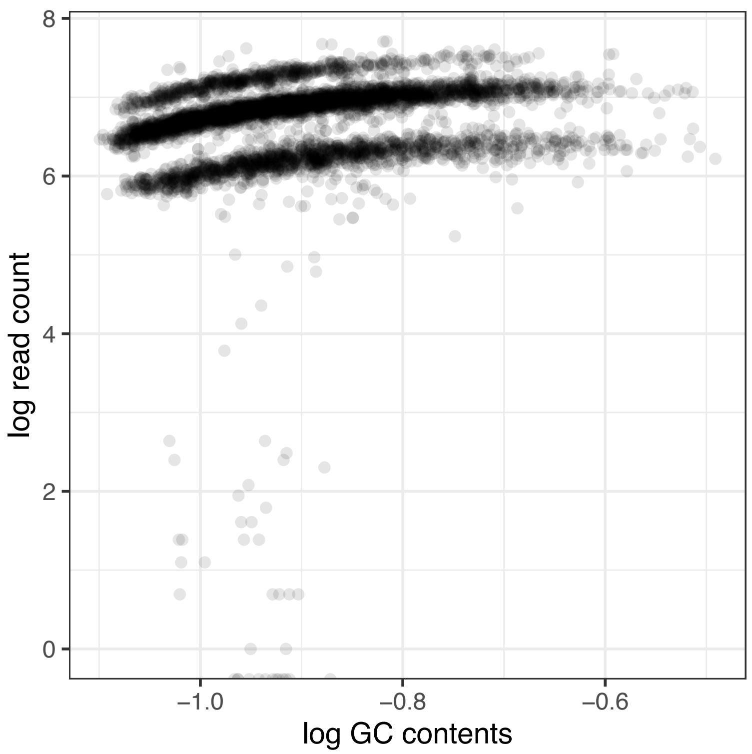
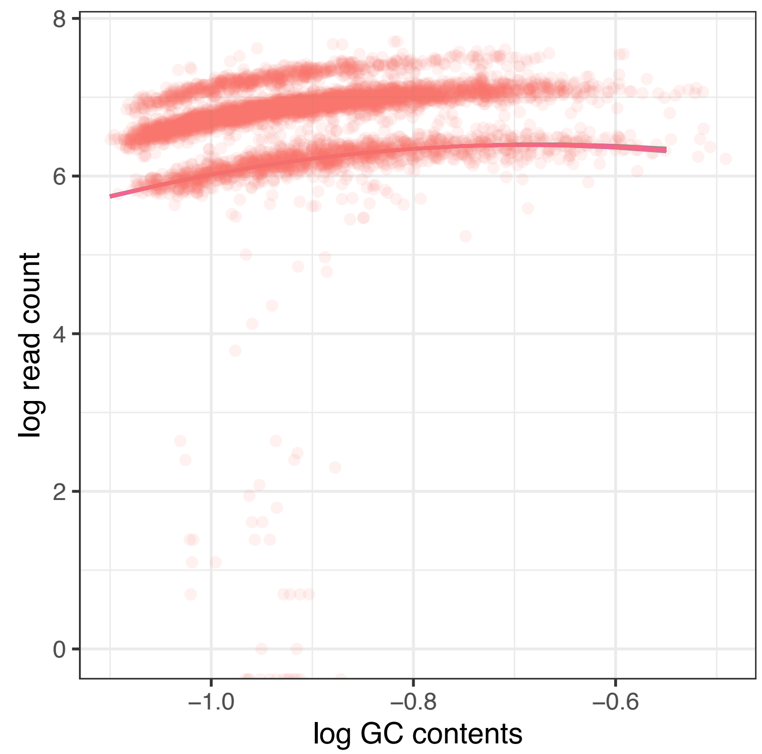
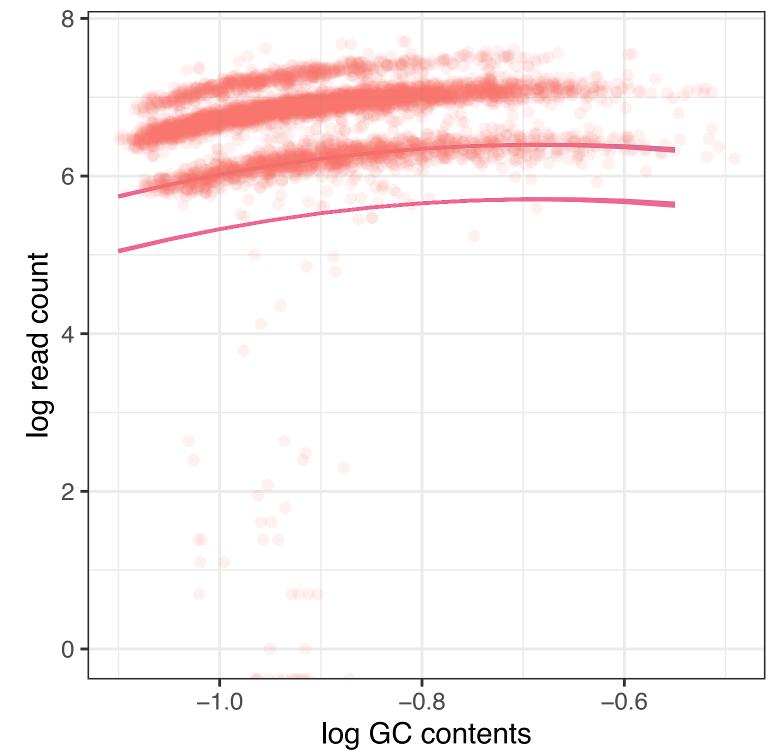
Data



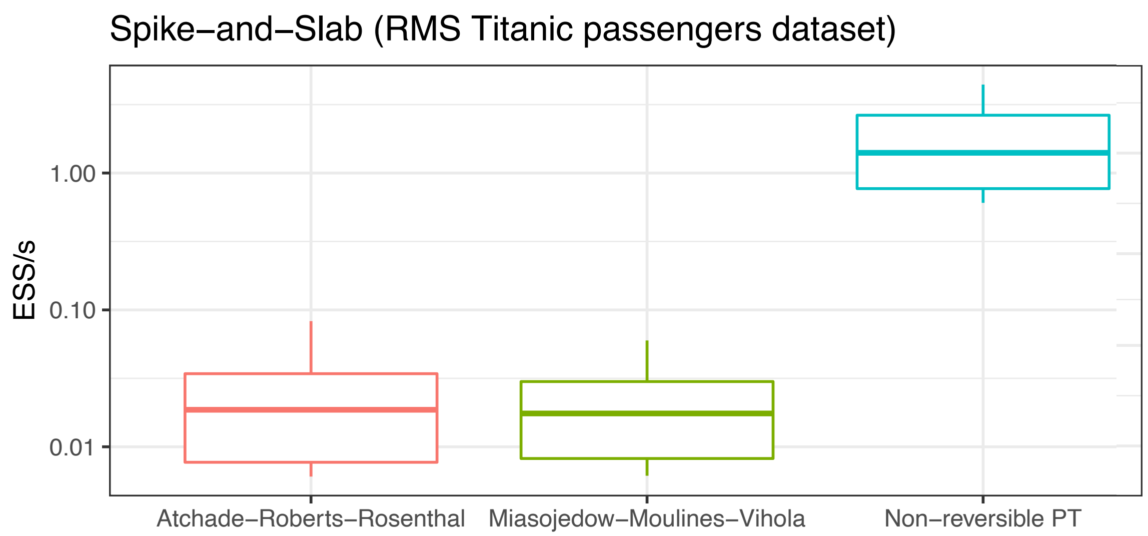
Single Chain

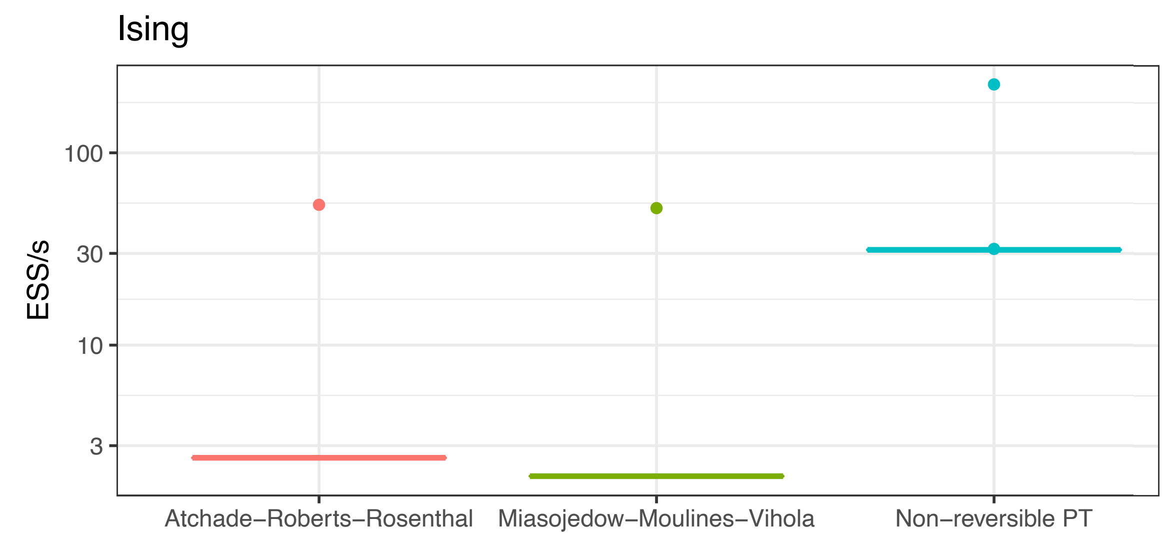


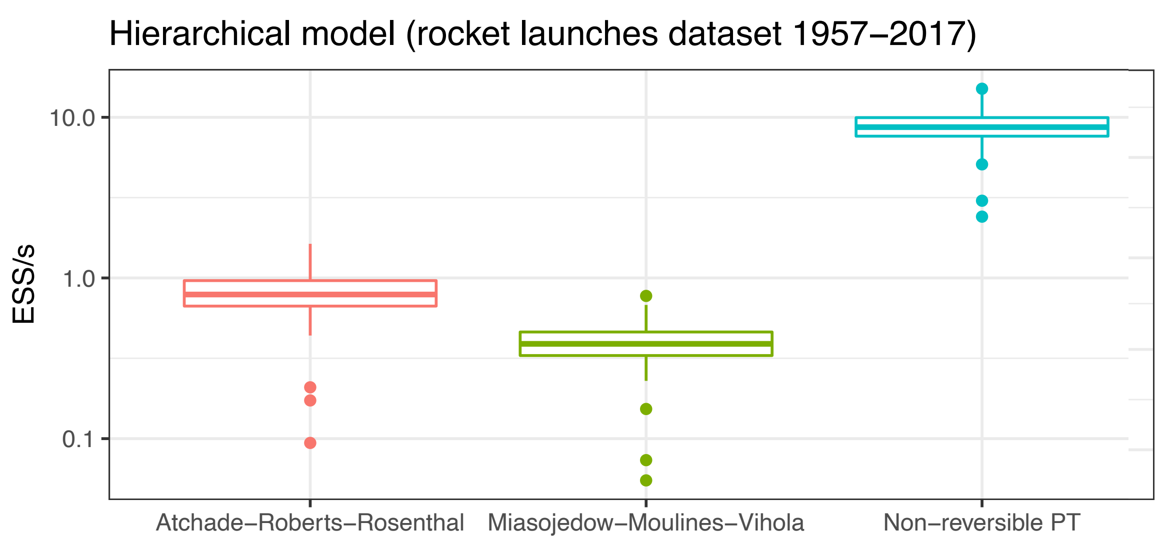
Non-Reversible PT

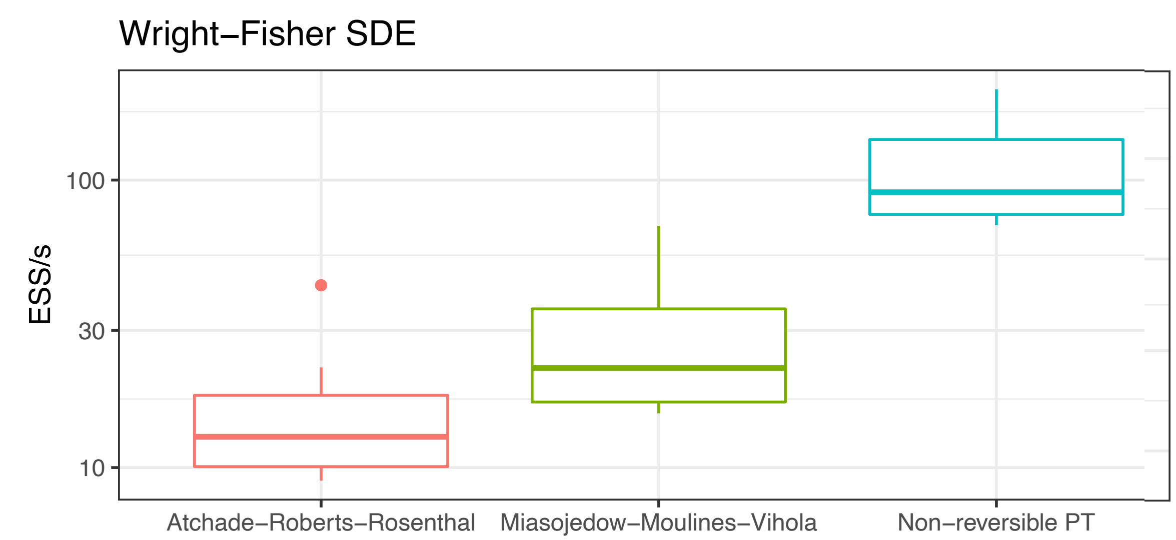


ESS COMPARED TO REVERSIBLE PT







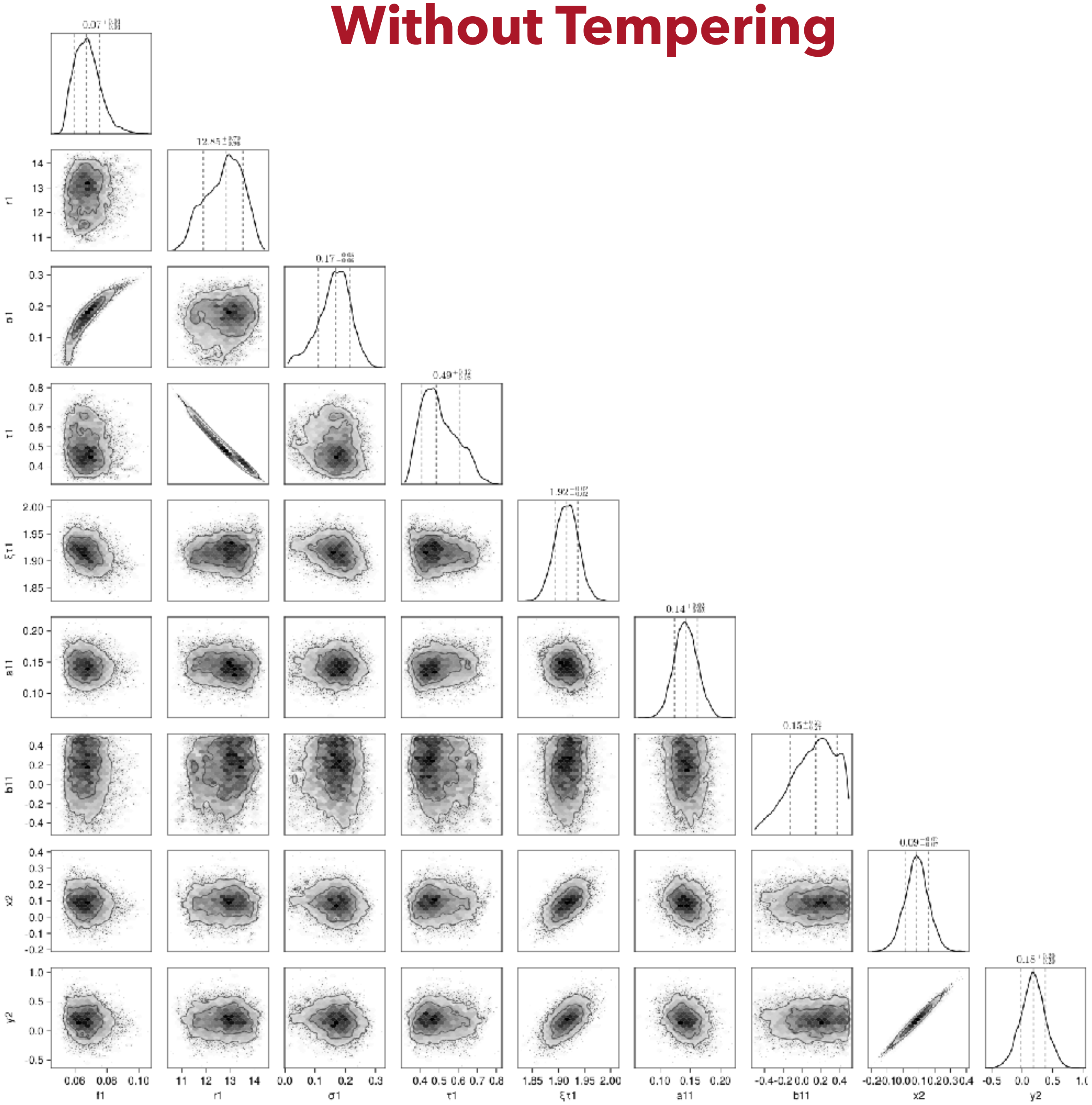


State of the art for Reversible PT

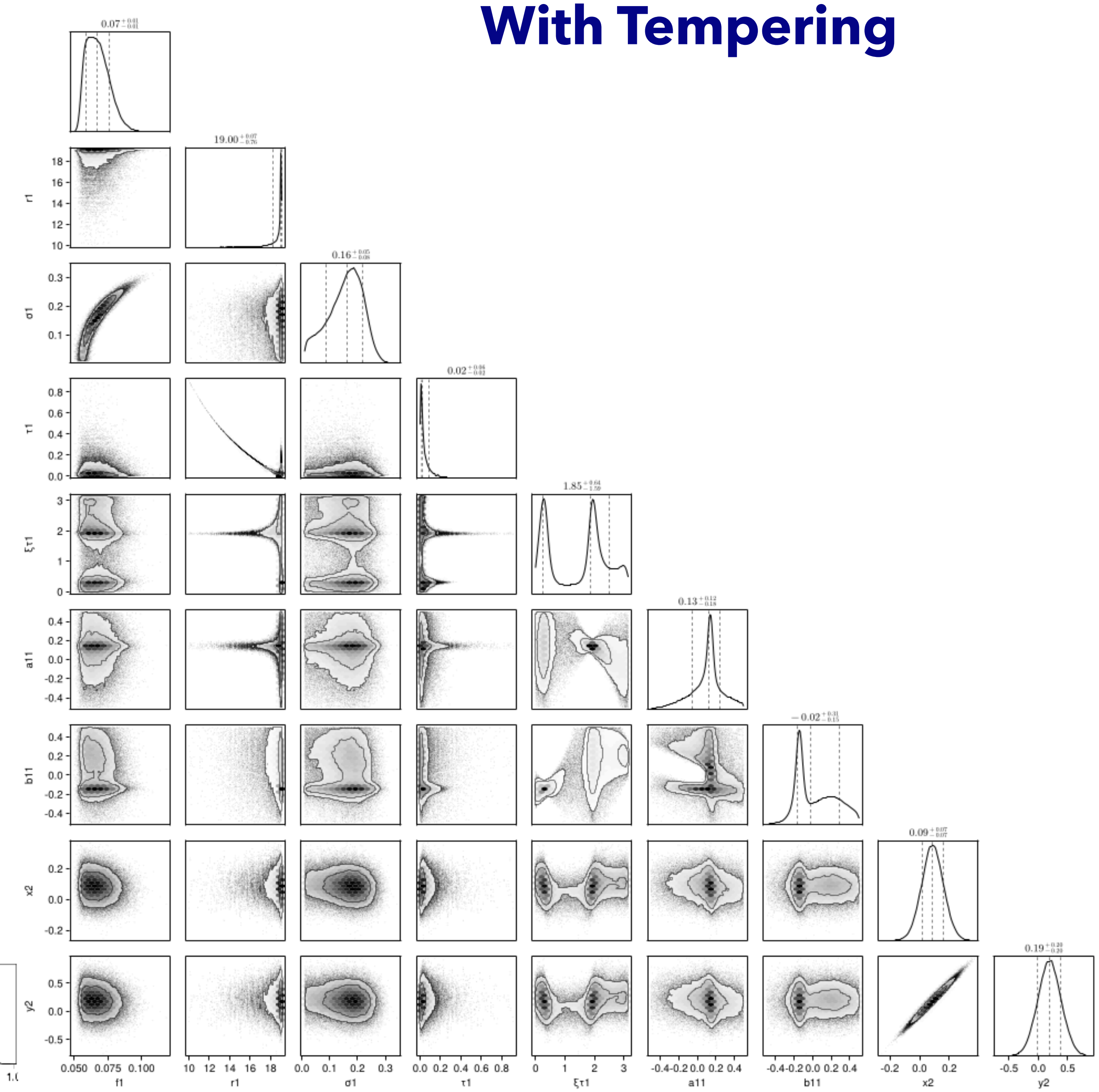
Non-Reversible PT

EXAMPLE: MULTIMODAL TARGETS IN ASTRONOMY

Without Tempering

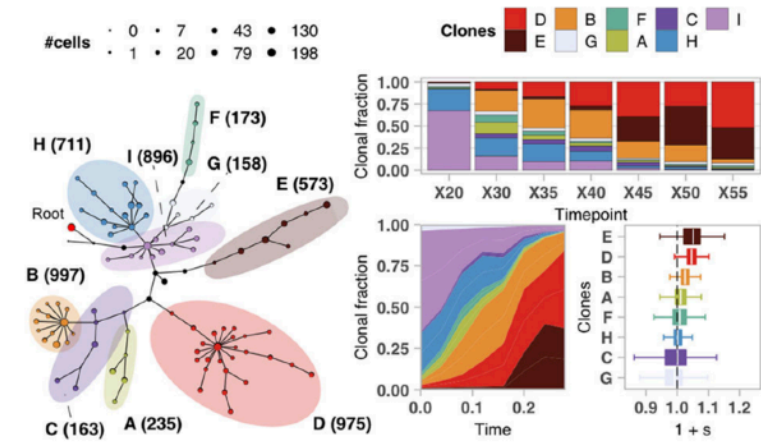


With Tempering



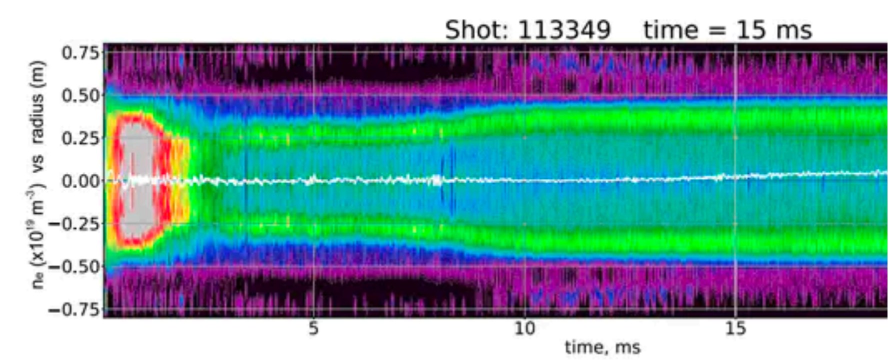
RECENT APPLICATIONS OF NRPT

- **Cancer:** Model evolutionary dynamics of single-cell cancer genomes.
  BC Cancer Research Institute & Sloan Kettering Cancer Center (Dorri et al. 2020, Salehi et al., 2021 in Nature)



- **Epidemiology:** Estimation of allele frequencies/study genetic variations in malaria.
  (Murphy & Greenhouse., 2024)

- **Nuclear Fusion:** Model plasma dynamics inside worlds largest FRC nuclear fusion reactor.
  Google Research & TAE Technologies (Gota et al., 2022, Langmore et al. 2023)



- **Political Science:** Identify gerrymandering in Georgia's 2021 congressional district plan
  Georgia Tech (Zhao et al., 2022, Chuang et al 2024)

- **Physics:** Detect and classify orbit of brown dwarfs
  (Jerry W. Xuan, 2024 in Nature)

- **Linguistics:** study the meanings of ancient Greek words change with time *(Zafar & Nicholls, 2024)*

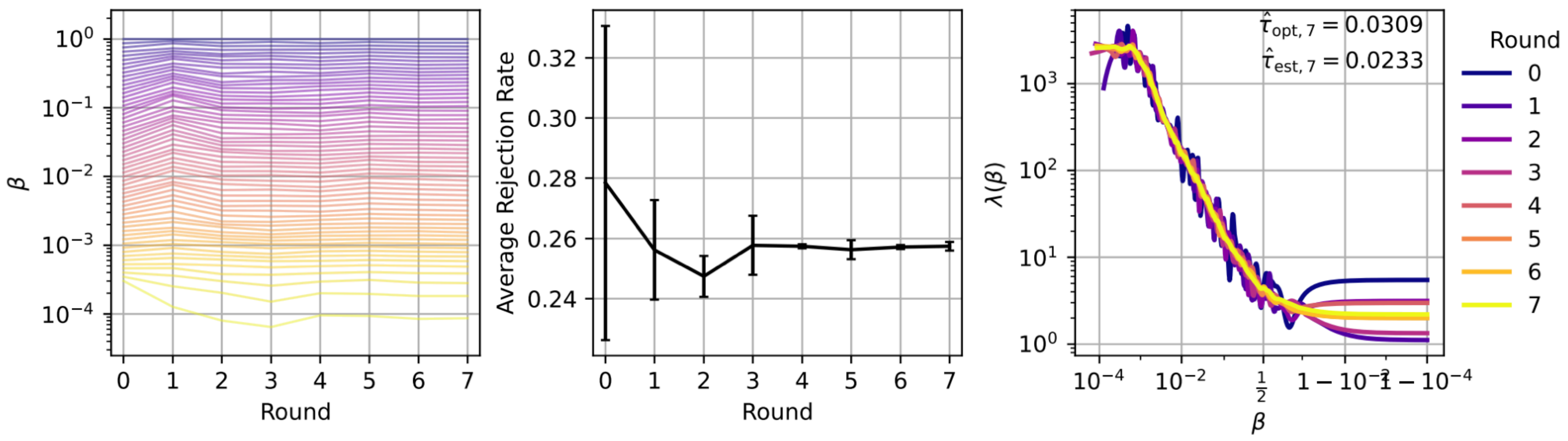
- **Probabilistic programming:** Inference engine for various PPL/Packages in different languages:
  Pigeons.jl, Blang, Comrade.jl, Themis, DrJacoby, MOIRE, Turing.jl, tensorflow probability, Hopsy

EVENT HORIZON TELESCOPE

- Event horizon telescope (EHT) used NRPT+HMC to more reliably recreate original photo of supermassive blackhole at M87 within 2% of computation budget
  Tiede, 2021

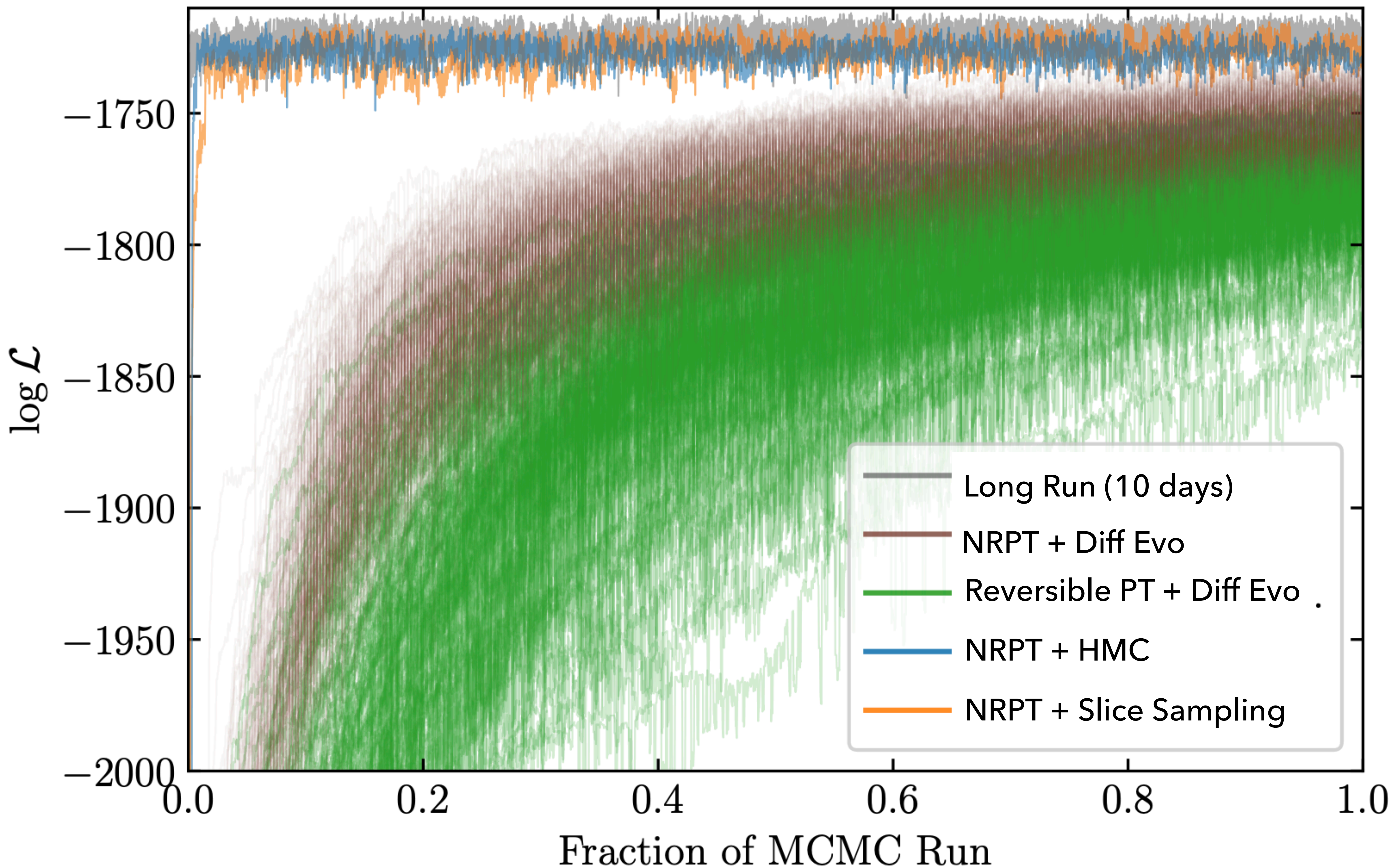
M87 (EHT 2019)

*Taken from Paul Tiede's PhD Thesis, 2021



NRPT FOR M87 BLACK HOLE PHOTO

*Taken from Paul Tiede's PhD Thesis, 2021



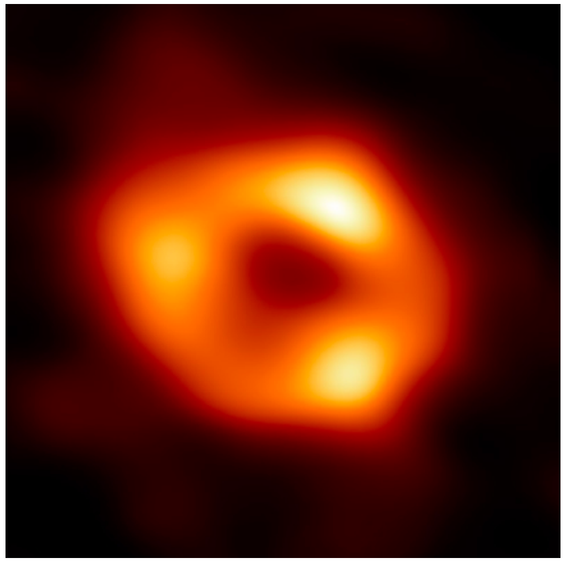
EVENT HORIZON TELESCOPE

EVENT HORIZON TELESCOPE

- Used to infer image of M87 and Sagittarius A* (EHT, 2022, 2024)

M87 (EHT 2019)

Sag A* (EHT 2022)



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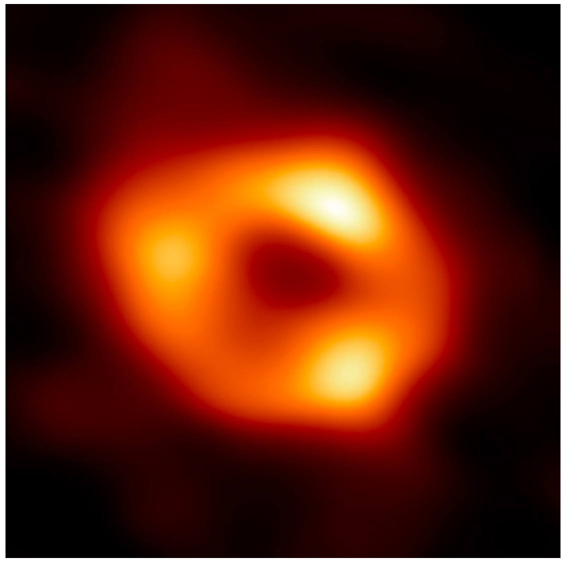
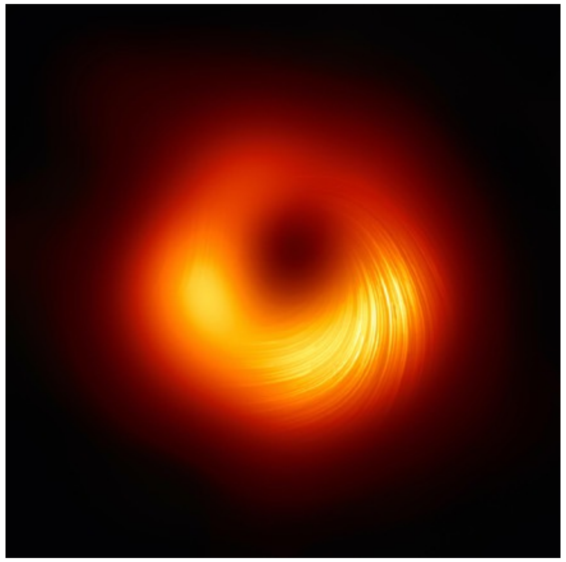


- Used to discover magnetic polarisation (EHT, 2021, 2024)

M87 (EHT 2021)



Sag A* (EHT 2024)